



VOYAGER 1 ENCOUNTERS SATURN



VOYAGER 1 ENCOUNTERS SATURN

The Project Gutenberg eBook of Voyager 1 Encounters Saturn

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Voyager 1 Encounters Saturn

Author: United States. National Aeronautics and Space Administration

Release date: December 18, 2017 [eBook #56205]

Most recently updated: October 23, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/56205

Credits: Produced by Stephen Hutcheson and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK VOYAGER 1
ENCOUNTERS SATURN ***

Voyager 1 Encounters Saturn

Contents

<u>Foreword</u>	1
<u>Introduction</u>	3
<u>The Planet</u>	4
<u>The Rings</u>	12
<u>The Satellites</u>	20
<u>A Glimpse Back</u>	32
<u>The Voyager Mission</u>	36
<u>MISSION OBJECTIVES</u>	36
<u>SPACECRAFT CHARACTERISTICS</u>	36
<u>SATURN ENCOUNTER</u>	36
<u>Scientific Highlights</u>	38
<u>SATURN</u>	38
<u>RINGS</u>	38
<u>NEW SATELLITES</u>	38
<u>INNER SATELLITES</u>	38
<u>TITAN</u>	39
<u>OUTER SATELLITES</u>	39
<u>MAGNETOSPHERE</u>	39
<u>Scientific Investigations</u>	40

A black and white photograph of Saturn and its moons Tethys and Dione against a starry background. Saturn is visible as a bright, slightly out-of-focus disk in the upper right. Tethys is a small, bright dot above Saturn, and Dione is a similar dot below it. A faint shadow of Tethys is visible on the cloudtops of Saturn in the upper right corner.

COVER: Saturn and two of its moons, Tethys (above) and Dione (below), were photographed by Voyager 1 on November 3, 1980, from 13 million kilometers (8 million miles). The shadow of Tethys is cast onto the cloudtops in the upper right corner of the image.

COVER: Saturn and two of its moons, Tethys (above) and Dione (below), were photographed by Voyager 1 on November 3, 1980, from 13 million kilometers (8 million miles). The shadow of Tethys is cast onto the cloudtops in the upper right corner of the image.

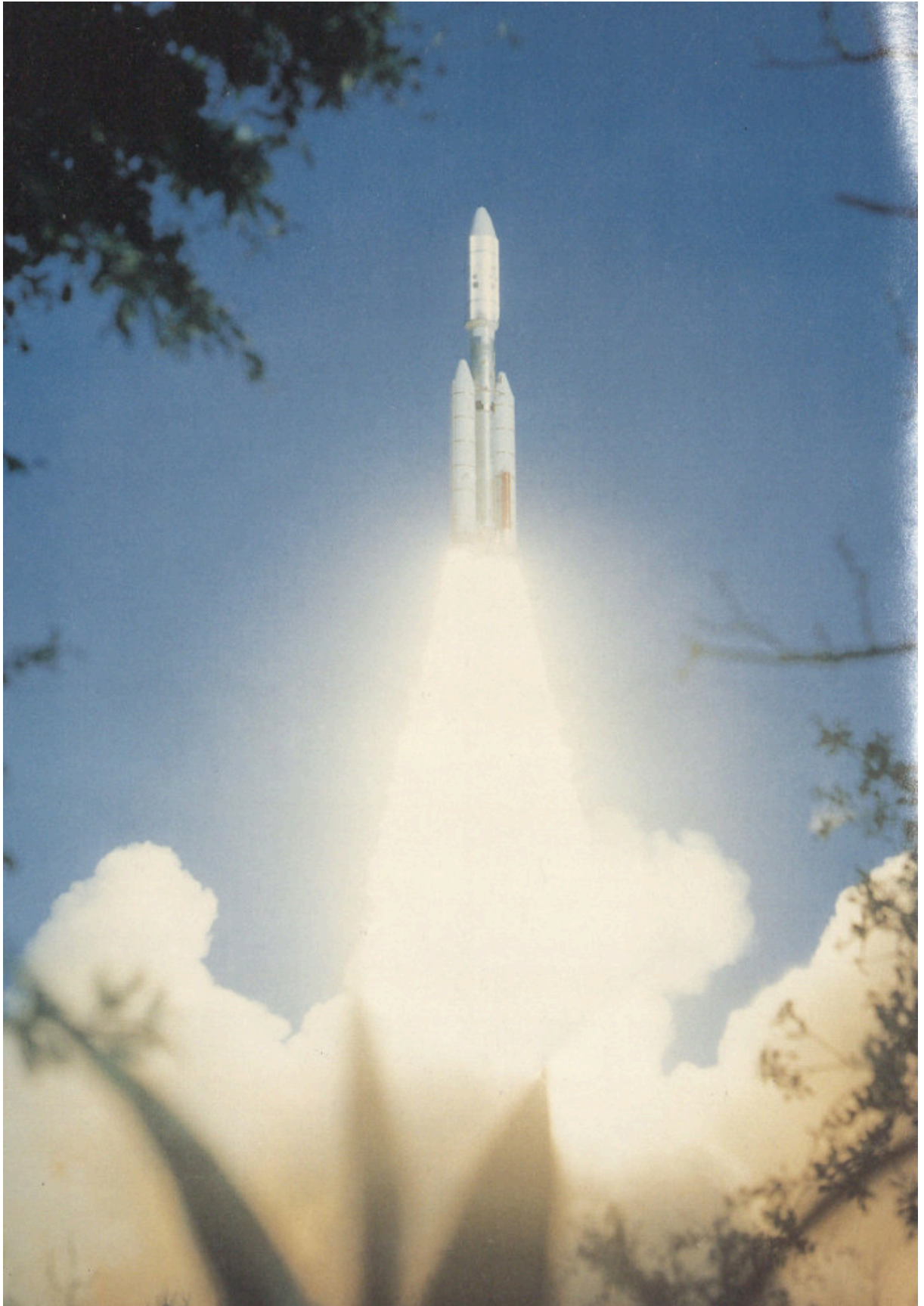
Foreword

The pictures assembled in this publication are a part of the rich and varied harvest of information returned by Voyager 1 across nearly a billion miles of interplanetary space. These images are of great beauty as well as great scientific interest, serving to remind us of the awesome and breathtaking dimensions of the solar system we inhabit. Voyager is providing intriguing new information which should help us to understand how the Earth—and possibly the universe—was formed. Already there have been surprises and puzzles that paint a completely new picture of Saturn and its neighborhood, including the discovery of three new moons, startling information about Saturn's rings, and observation of the unexpectedly complex structure of Saturn's atmosphere and that of its largest moon, Titan. It will take years for scientists to assimilate completely the information which is cascading down from Voyager. What more will this marvel of technology have to tell us before it departs the solar system to travel endlessly among the stars?

ROBERT A. FROSCH, *Administrator*
National Aeronautics and Space Administration
December 1980

The date of each photograph and the distance of the spacecraft from the planet or satellite are included with each picture.

For sale by the Superintendent of Documents, U.S. Government
Printing Office, Washington, D.C. 20402
Stock No. 033-000-00817-1



Voyager 1 was launched from Cape Canaveral, Florida, on September 5, 1977, beginning its journey to Jupiter, Saturn, and beyond.

Introduction

No other generation has had the opportunity or the technology to reach beyond our world—to see, to touch, to hear the forces that shape our universe. In slightly over two decades, man has ingeniously explored five distant planets—and two dozen moons. We have seen their weather and surfaces, landed on some, probed the atmospheres of others, and listened to their radio noises.

Under the planetary exploration program of the National Aeronautics and Space Administration, the Voyager Mission, begun in 1972, was designed to explore Jupiter, Saturn, their satellites, rings, magnetic fields, and interplanetary space. Two automated, reprogrammable spacecraft, Voyagers 1 and 2, were launched in late summer of 1977. Their goals: the outer planets.

Both spacecraft made astounding discoveries in the Jupiter system in 1979—a thin ring, a thick ionized sulfur and oxygen torus, an actively volcanic satellite—these were but a few of the treasures yielded by the two Jupiter flybys.

Now, Voyager 1 has completed exploration of its final target: the ringed planet Saturn and its enigmatic giant satellite, Titan. True to the generally unpredictable nature of planetary exploration, the treasures of the Saturn system far exceeded all expectations. We learned more about Saturn in one week than in all of recorded history, thanks to one trusty robot no larger than a compact car and to thousands of diligent and imaginative people.

Both spacecraft carry an assortment of optical, radiometric, and fields and particles sensing instruments. Taken together, their data present a comprehensive picture of a planetary system—and clues to what is happening, what has happened, and what may happen in our universe.

This publication presents the preliminary photographic results of Voyager 1's encounter with Saturn and its major satellites. Voyager 1 transmitted over 17,500 images in its four months of close observations of the system. Many of these images have been combined to produce mosaics and color pictures. Hundreds have yet to be closely examined.

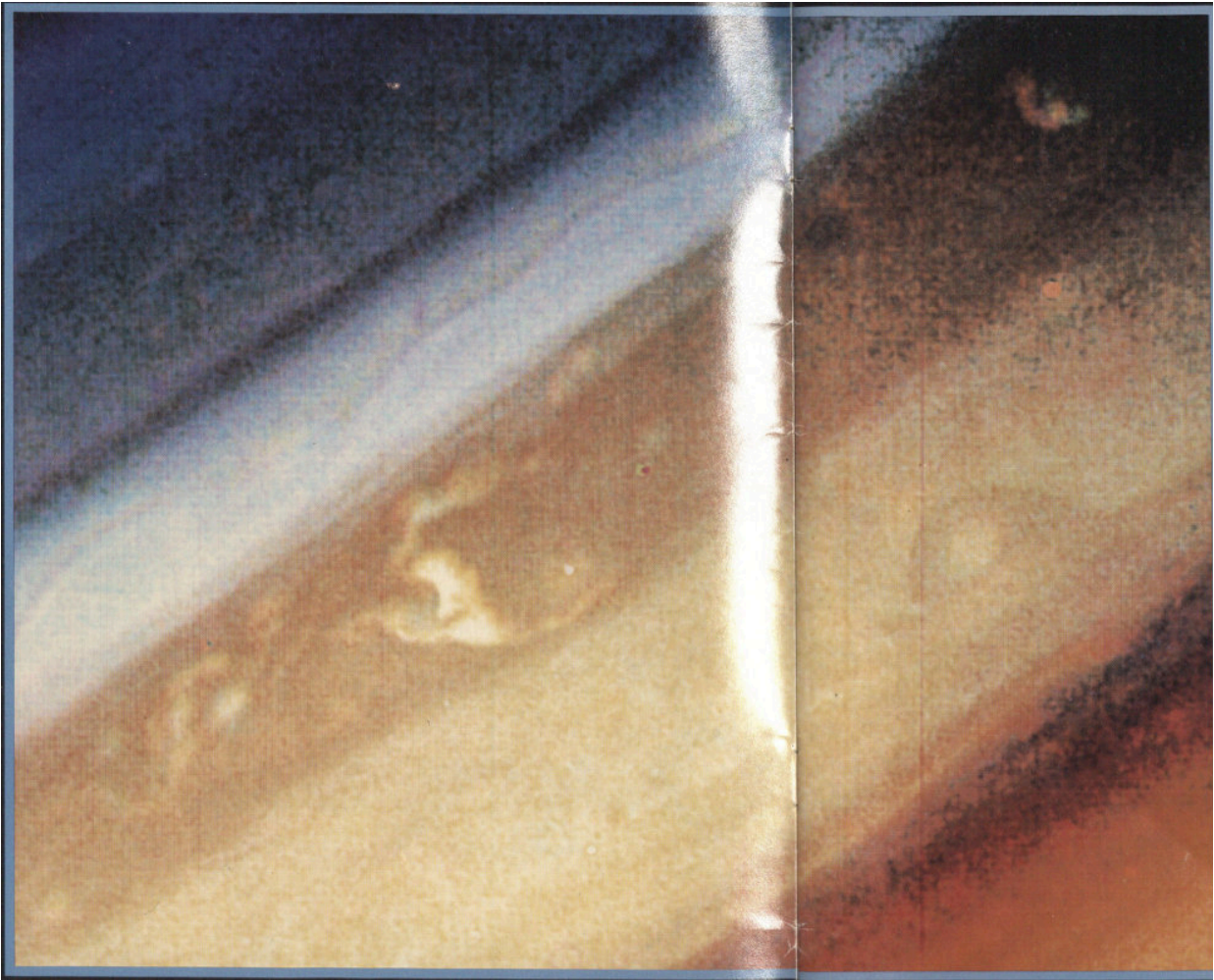
The second Voyager spacecraft will begin its close Saturn observations in early June 1981 and make its closest approach to the planet's northern hemisphere on August 25. Then, due to its launch during a period of rare planetary alignment occurring only once every 175 years, Voyager 2 will be able to continue on to a rendezvous with the seventh planet, Uranus, in January 1986, and perhaps even the eighth planet, Neptune, in August 1989.

Voyager 1's primary mission is complete. But its usefulness is far from over. As we go about our daily business, Voyager 1 is searching for another frontier—the edge of our solar system. In 7 to 15 years, the spacecraft will cross the heliopause—the farthest reaches of our Sun's magnetic field influence. Then, high above our ecliptic plane, Voyager 1 will continue its flight toward the star Alpha Ophiuchus. Eventually, Voyager 1 will be too distant to communicate with Earth and will silently drift in space forever.

ANDREW J. STOFAN, *Acting Associate Administrator for Space Science
National Aeronautics and Space Administration*

The Planet

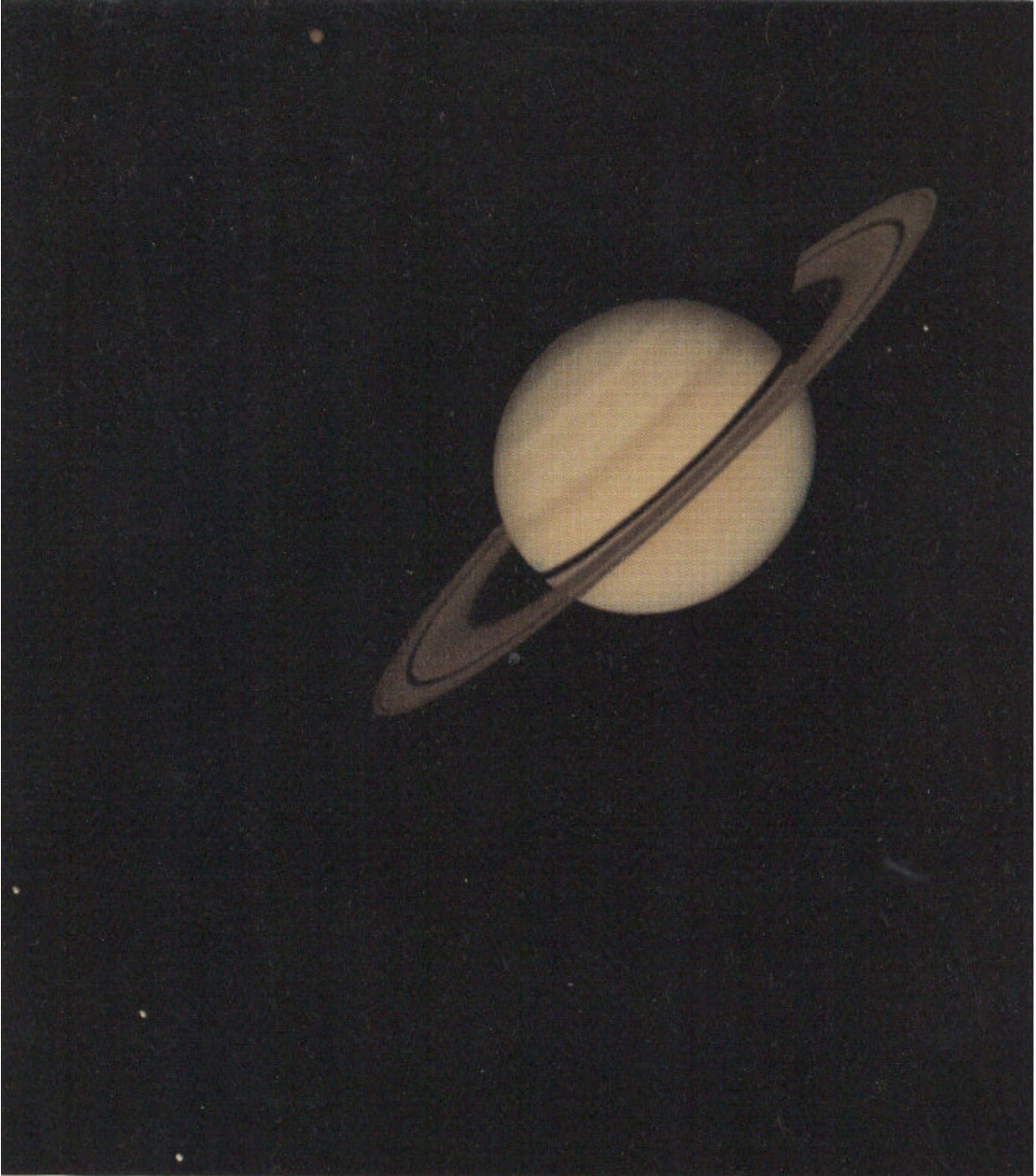
5



11/5/80 9 million km (5.5 million mi)

Saturn is the sixth planet from the Sun and second largest in our solar system. Like Jupiter, it is a giant sphere of gas—mostly hydrogen and helium—with a possible core of rocky material. Various features in Saturn's cloudtops are visible in the accompanying color-enhanced image of the planet's northern hemisphere: small-scale convective cloud features (similar to, but much larger than, thunderstorms in Earth's atmosphere) are visible in the brown belt; an isolated convective cloud with a dark ring is visible in the light brown zone; and a longitudinal wave is visible in the light blue region.

6

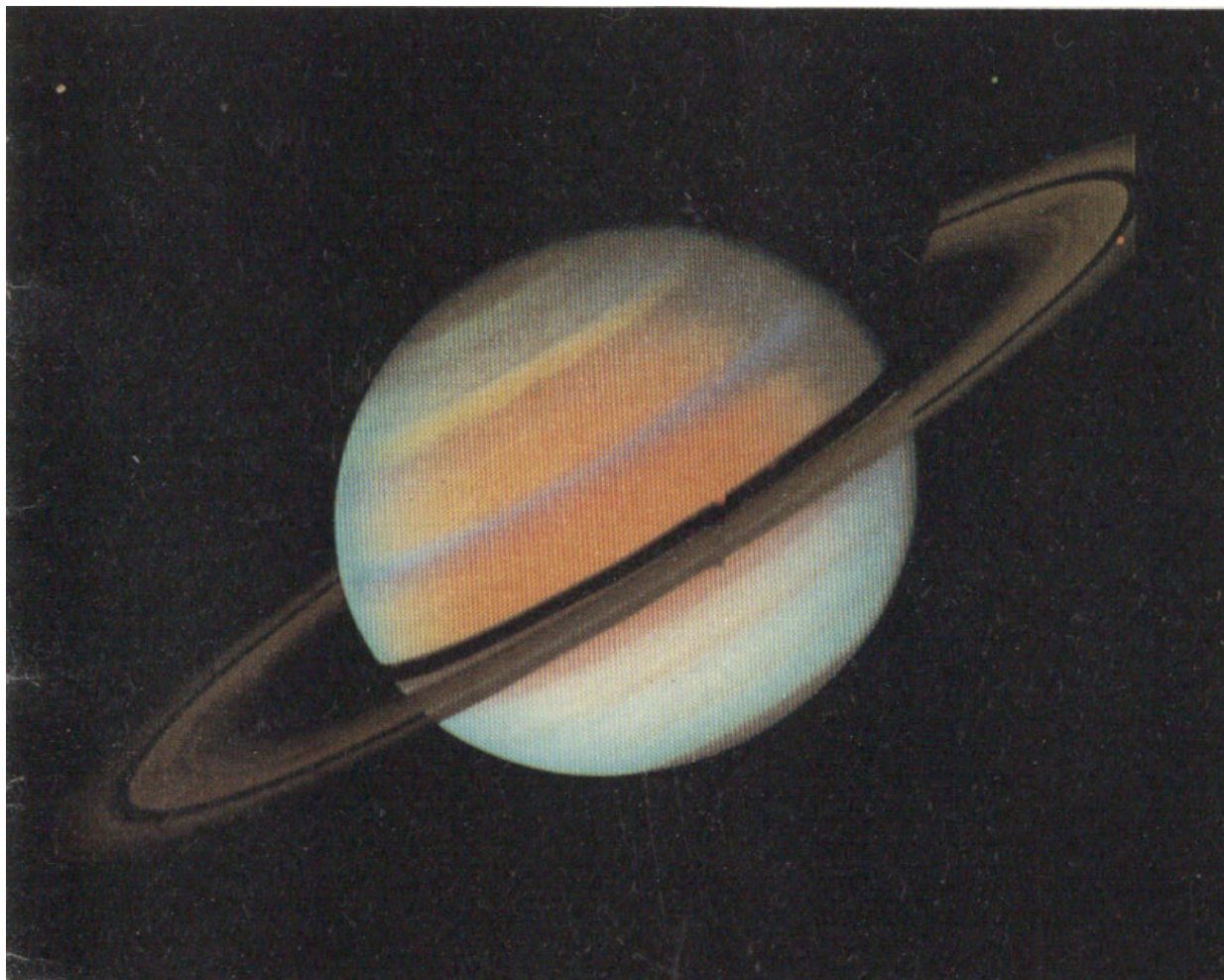


9/17/80 76 million km (47 million mi)

As Voyager 1 approached Saturn, a series of dark and light cloud bands (belts and zones) became apparent in the planet's northern hemisphere through a high altitude atmospheric haze. The planet's shadow obscures the rings behind and immediately to the east of the disk. In addition, the shadow of the rings on the planet's disk can be seen just north of the rings themselves as they cross in front of the planet. Six of Saturn's 15 known satellites are visible. Saturn's largest moon, Titan (considerably larger than Earth's moon), is clearly visible in the upper left corner. The smaller satellites Dione, Tethys, and Rhea are shown in the lower left corner (upper, middle, and lower, respectively). Two of the innermost moons, Mimas and Enceladus, appear to the right of the planet (Mimas is the one closer

to the planet). These six moons orbit Saturn in the equatorial plane and appear in their present positions because Voyager is above that plane.

7



10/18/80 34 million km (21 million mi)

The North Temperate Belt is visible as the violet-colored belt in this false-color photograph. In this image, features which are especially bright in ultraviolet light appear as turquoise and violet, while ultraviolet-dark areas appear orange. Notice in particular the three spots (two bright orange and one pale violet) at mid-northern latitudes. The bright spots are similar to those shown at much higher resolution in later images. The distinct color difference between the North Equatorial Belt and Saturn's other belts and zones may be due to a thick haze layer covering the northern portion of the belt. It is not yet understood why the southern hemisphere of the planet (below the rings) appears bluer than the northern hemisphere. Color spots in the rings are artifacts of image processing.



10/30/80 18 million km (11 million mi)

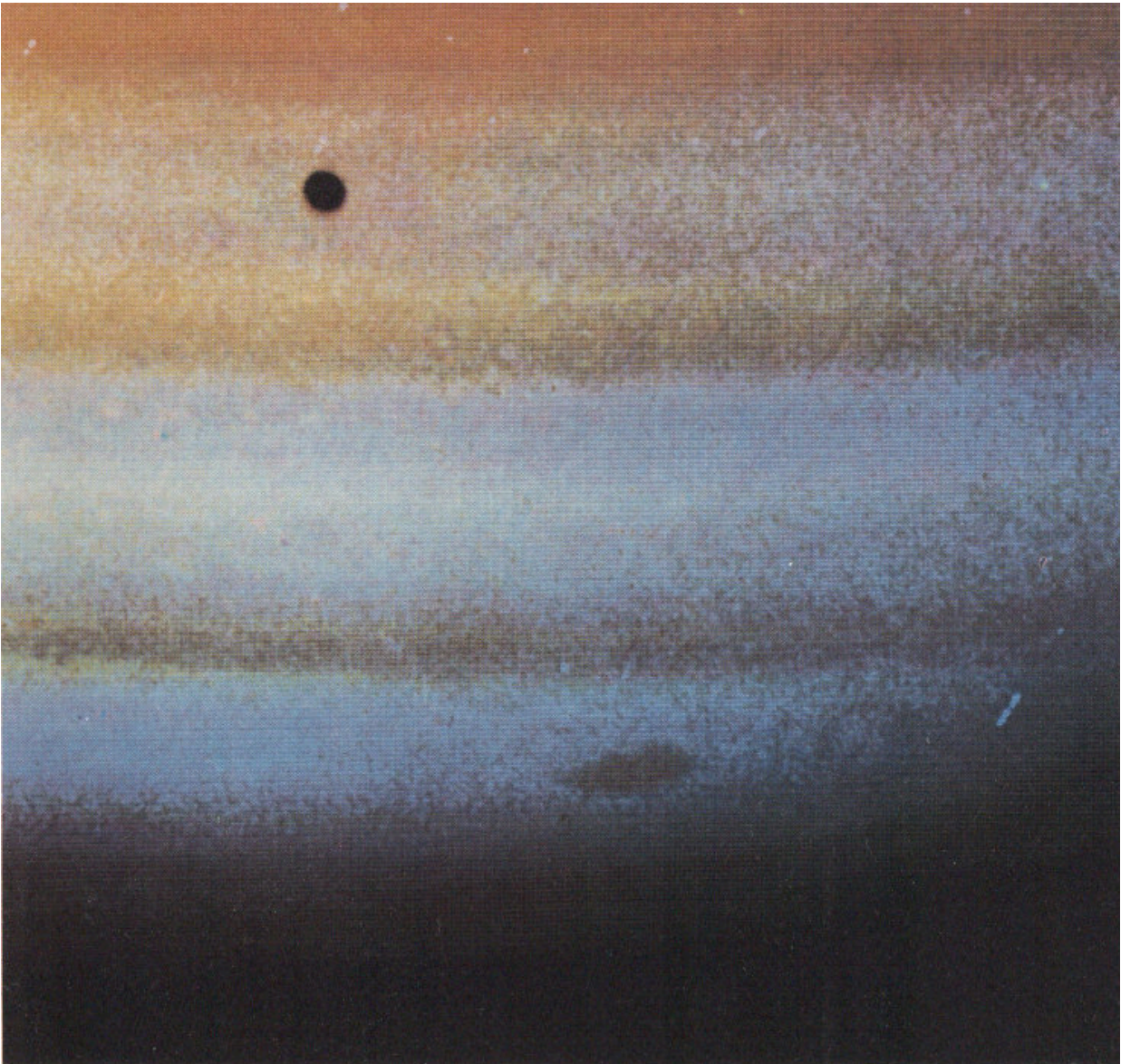
Saturn's soft, velvety appearance and previously unseen detail in its mysterious rings became visible as Voyager 1 approached the planet. For example, a gap in the dark C-Ring is now visible, and material can be seen within the relatively wide Cassini Division (long believed to be empty), which separates the B-Ring (middle) from the A-Ring (outer). The Encke Division appears near the outer edge of the A-Ring. Detail can be seen within the shadow cast by the rings upon the planet: the broad, dark band near the equator is the shadow of the B-Ring; the thinner, brighter line just to the south is the shadow of the less dense A-Ring. Three of Saturn's moons, Tethys (outer left), Enceladus (inner left), and Mimas (right) are also visible in this computer mosaic of Voyager 1 images.



11/6/80 8.5 million km (5.3 million mi)

An unusual red oval cloud feature, similar to (but smaller than) Jupiter's Great Red Spot, was discovered in the southern hemisphere of Saturn. The oval, 6000 kilometers (4000 miles) in length, is located at 55 degrees south latitude. The difference in color between the red oval and the surrounding bluish clouds in these two false-color images indicates that material within the oval contains a substance that absorbs more blue and violet light than the bluish clouds. Voyager scientists first observed the oval in August 1980, and the feature has retained its appearance since its discovery.

9



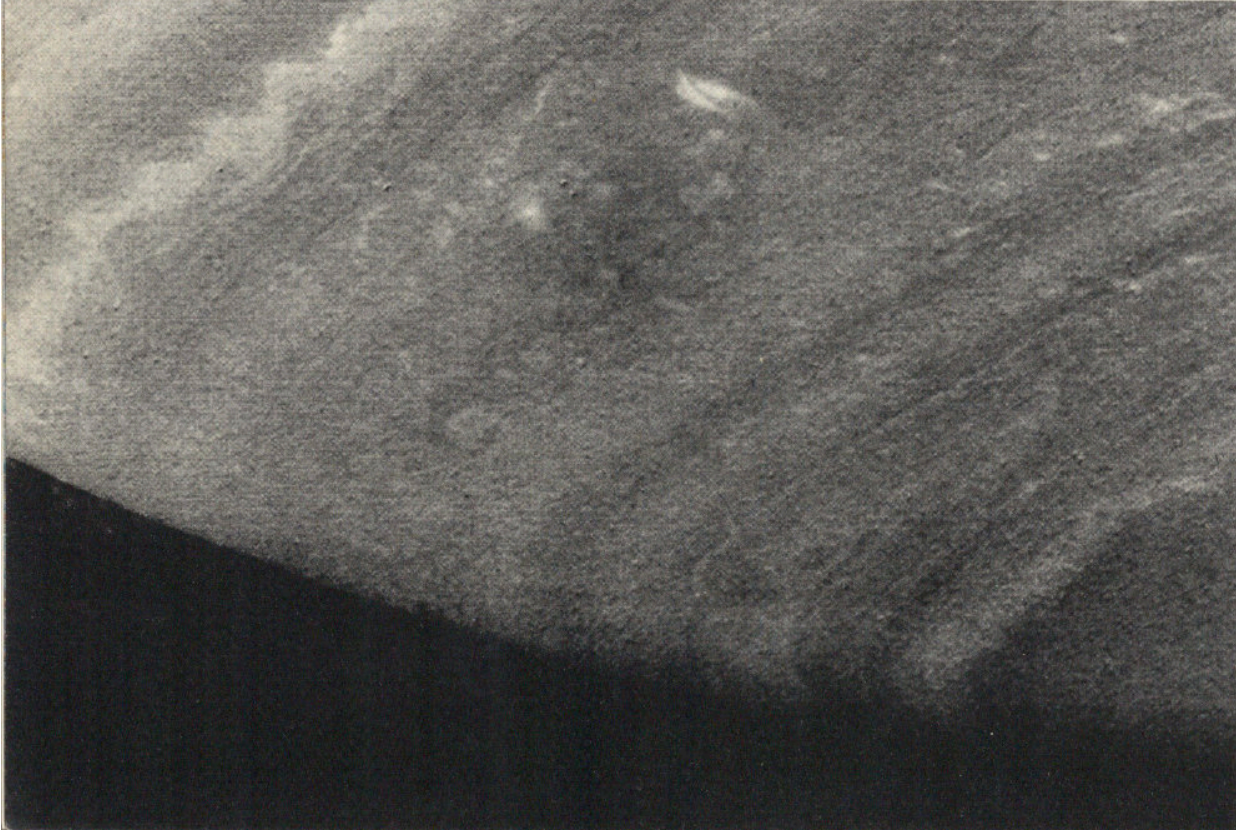
11/6/80 8 million km (5 million mi)

In this photograph, the shadow of the satellite Dione is seen as a dark circle on the face of the planet.

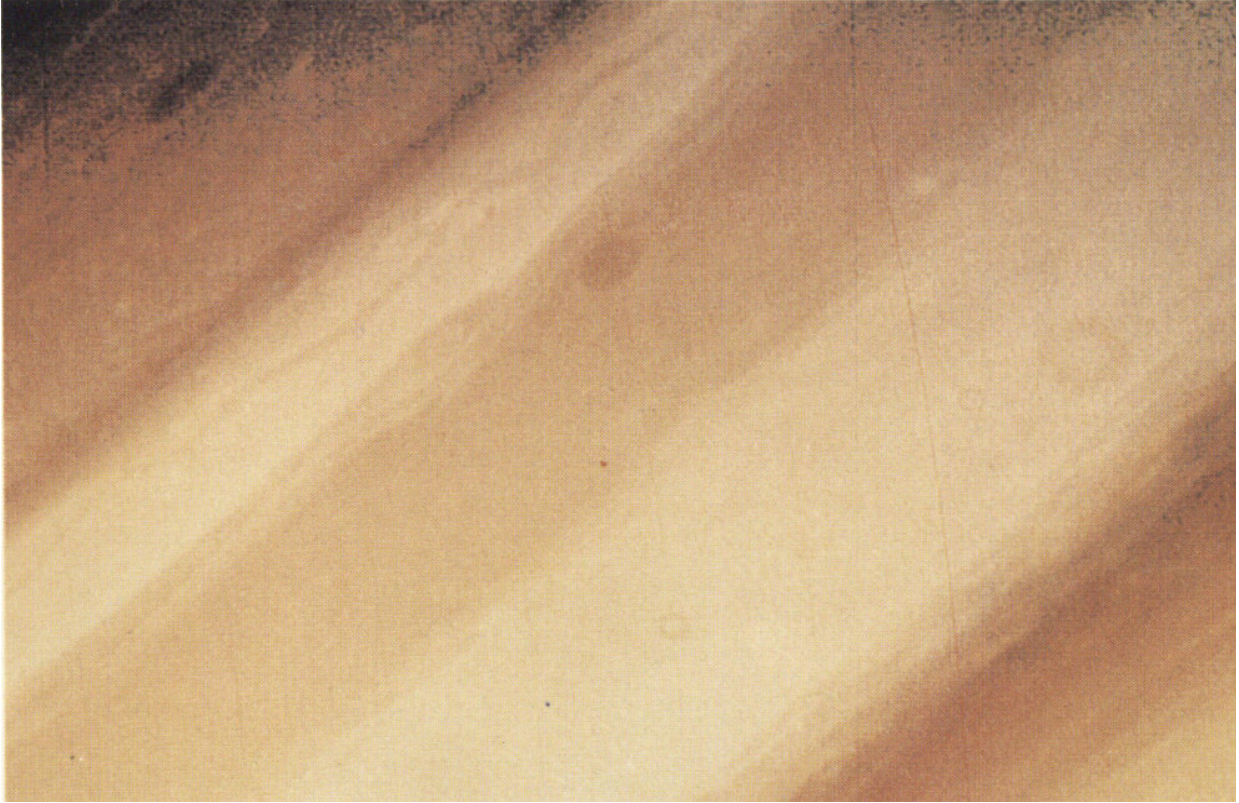


11/10/80 3.5 million km (2.2 million mi)

A ribbon-like wave structure and small convective features marking a westward jet stream above the wave are visible in this photograph of Saturn's cloudtops. The view, extending from 40 degrees to 60 degrees north latitude, shows features 65 kilometers (40 miles) in diameter. Measurements in images such as this one indicate that Saturn has fewer east-to-west wind currents than does Jupiter.



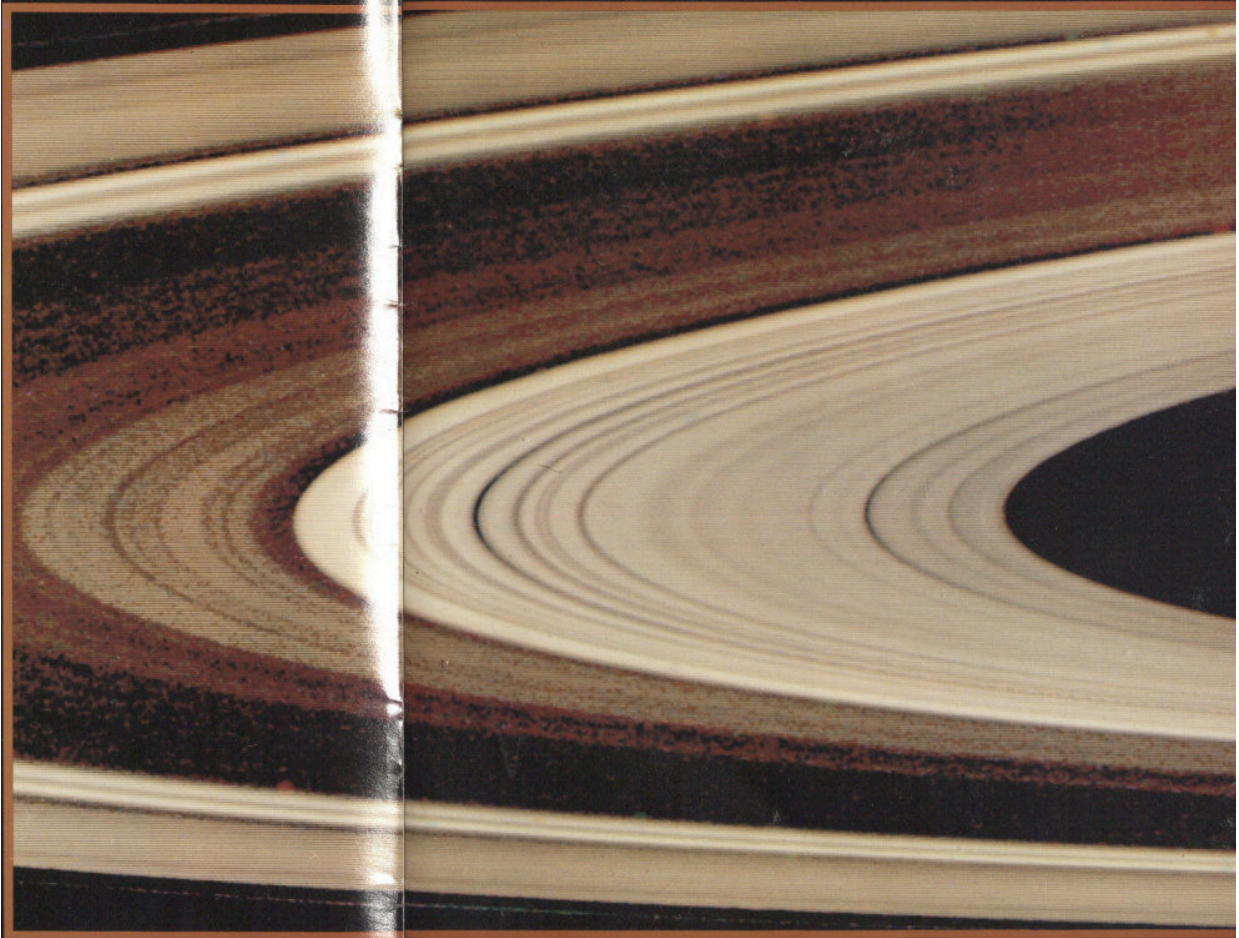
11/12/80 442,000 km (265,000 mi)
Numerous small cloud features were photographed as Voyager 1 passed above Saturn's southern hemisphere. At these polar latitudes, the large-scale light and dark bands break down into small-scale features, seen here as waves and eddies.



11/7/80 7.5 million km (4.6 million mi)

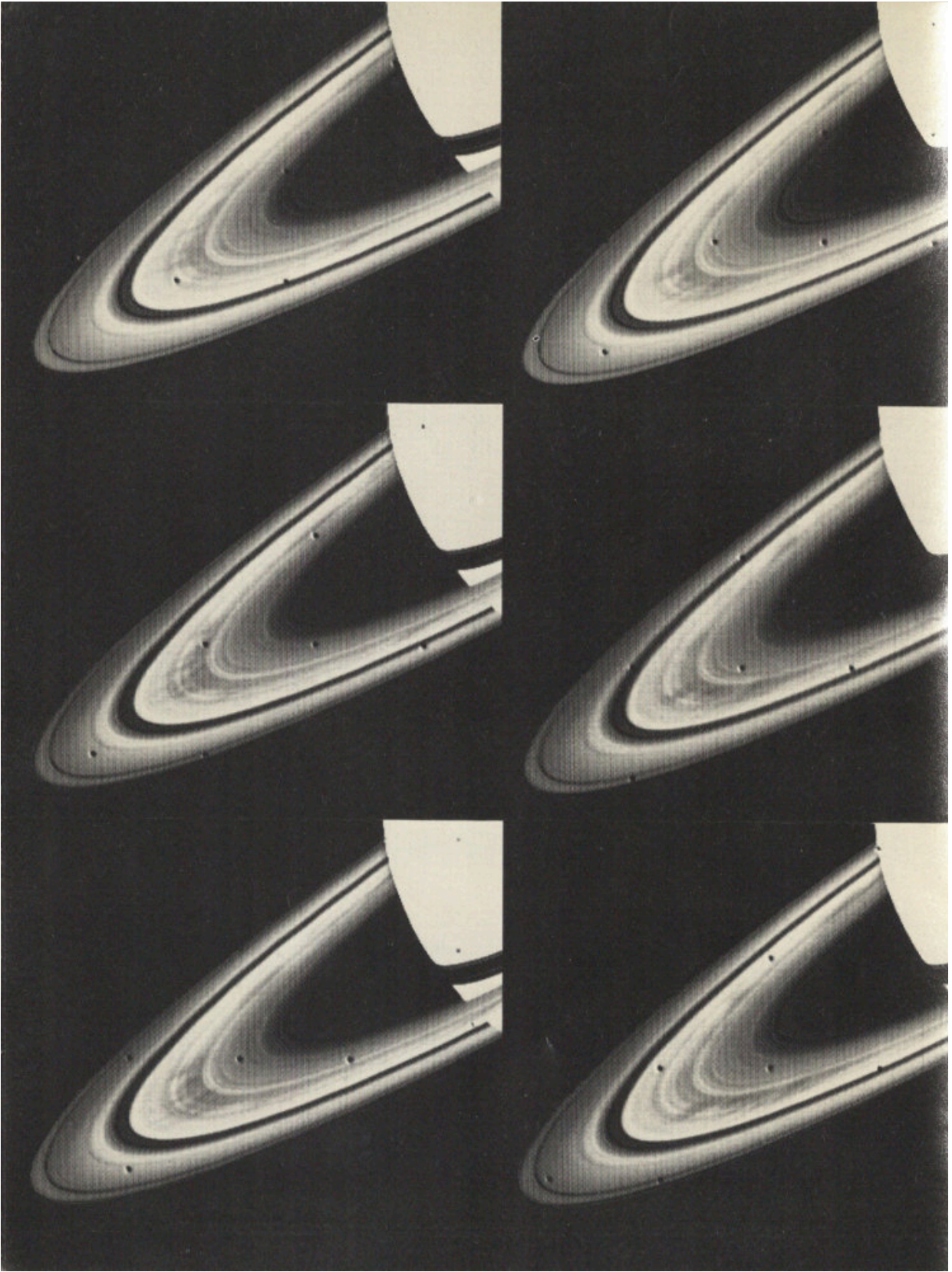
Two brown ovals, approximately 10,000 kilometers (6000 miles) across, were discovered in Saturn's northern hemisphere, at about 40 degrees and 60 degrees latitude. The polar oval (upper left) has a structure similar to Saturn's red oval located in the southern polar latitudes. Detail within the ovals is not visible at this resolution, so it is not yet known if they are rotating features similar to the many spots in Jupiter's atmosphere.

The Rings



11/12/80 717,000 km (444,000 mi)

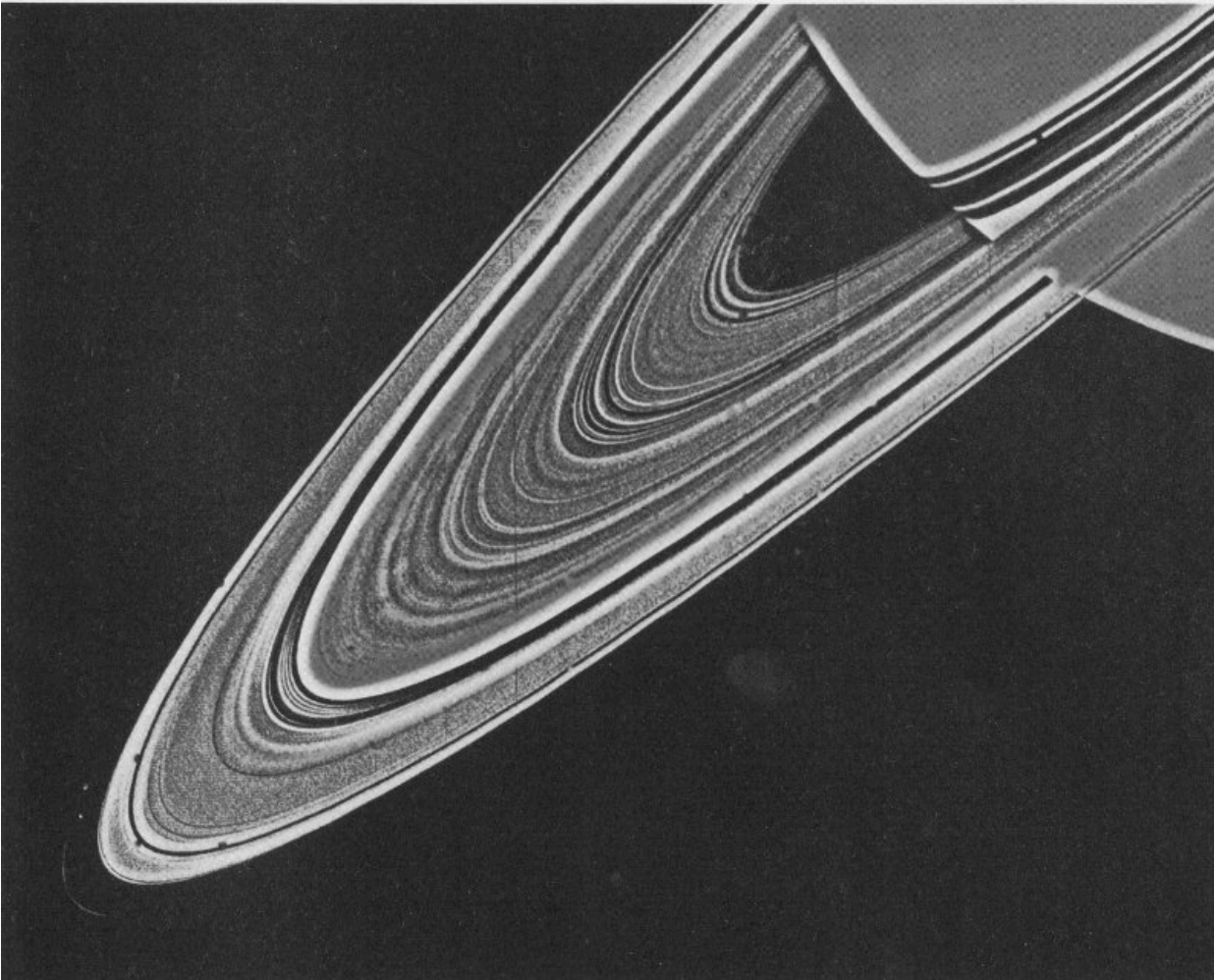
The rings of Saturn have amazed and intrigued astronomers for over 300 years. Now that we have seen them up close, they are even more astonishing. Although they stretch over 65,000 kilometers (40,000 miles), they may be only a few kilometers thick. The ring particles—from a few microns to a meter (three feet) in size—have been described as icy snowballs or ice-covered rock. Voyager scientists continue to pore over their data, searching for answers to the puzzles of the rings. The rings were named in order of their discovery, so the labels do not indicate their relative positions. From the planet outward, they are known as D, C, B, A, F, and E.



10/25/80 24 million km (15 million mi)

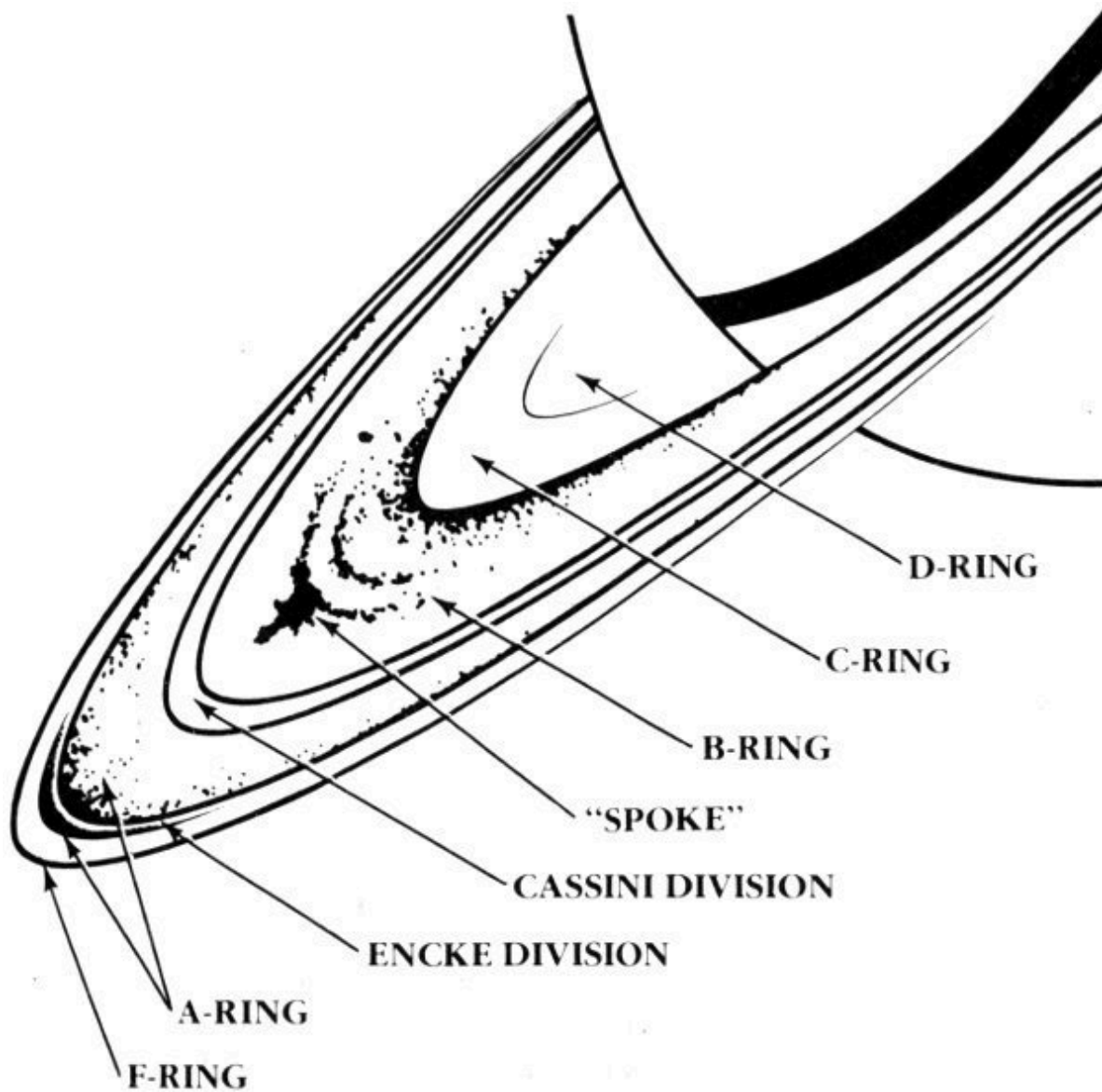
Extraordinarily complex structure is seen across the entire span of Saturn's ring system. The sequence (taken approximately every 15 minutes as Voyager 1 approached Saturn) proceeds from top to bottom in each column and shows radial "spokes" rotating within the B-Ring. The spokes may be caused by a combination of magnetic and electrostatic forces.

15



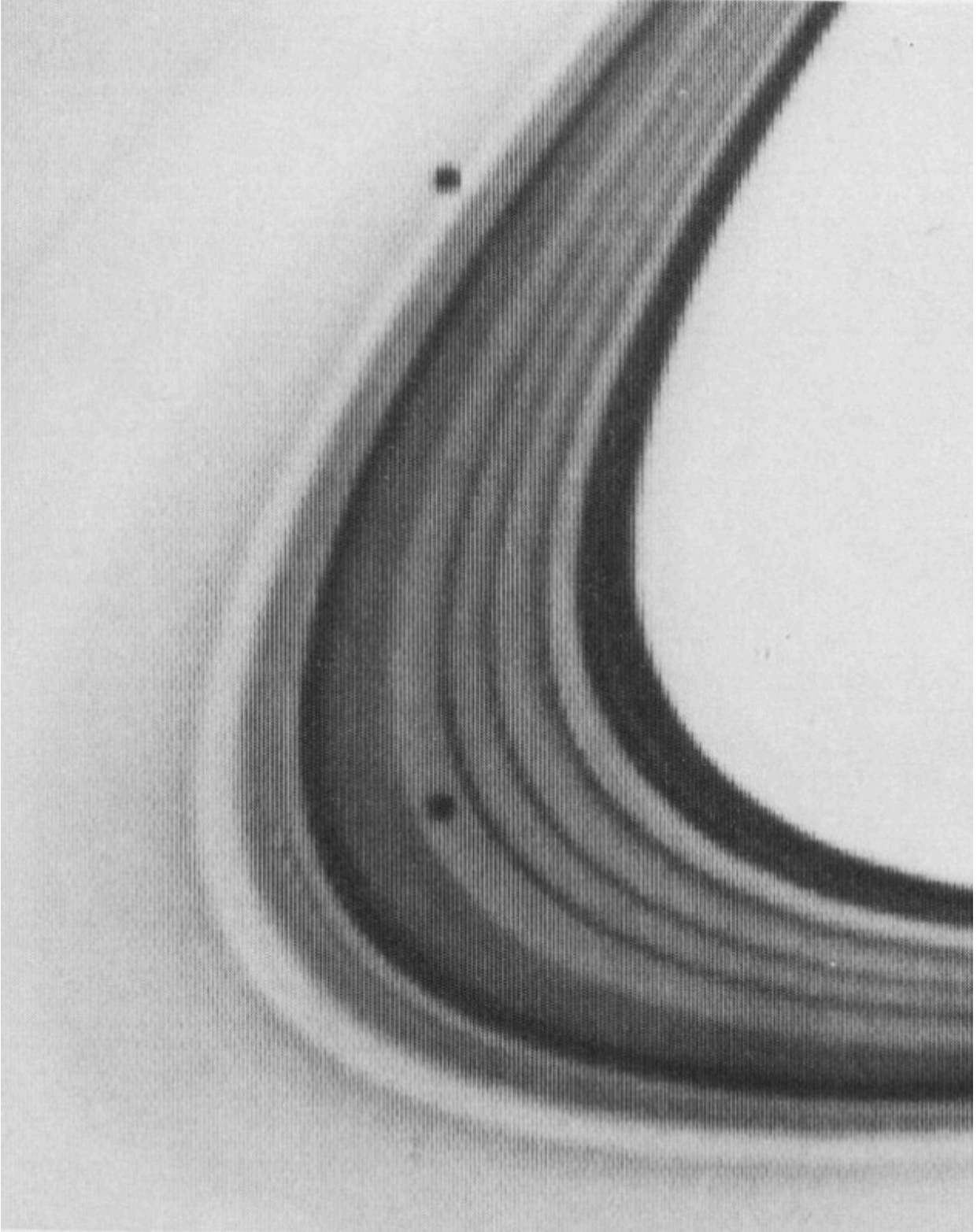
11/6/80 8 million km (5 million mi)

Over 95 individual concentric features can be counted; the final count in higher resolution images may be anywhere from 500 to 1000 separate rings. A few of the ringlets shown in this computer-assembled mosaic are not concentric circles but are instead elliptical. Ring particles are probably ice or ice covered rock.

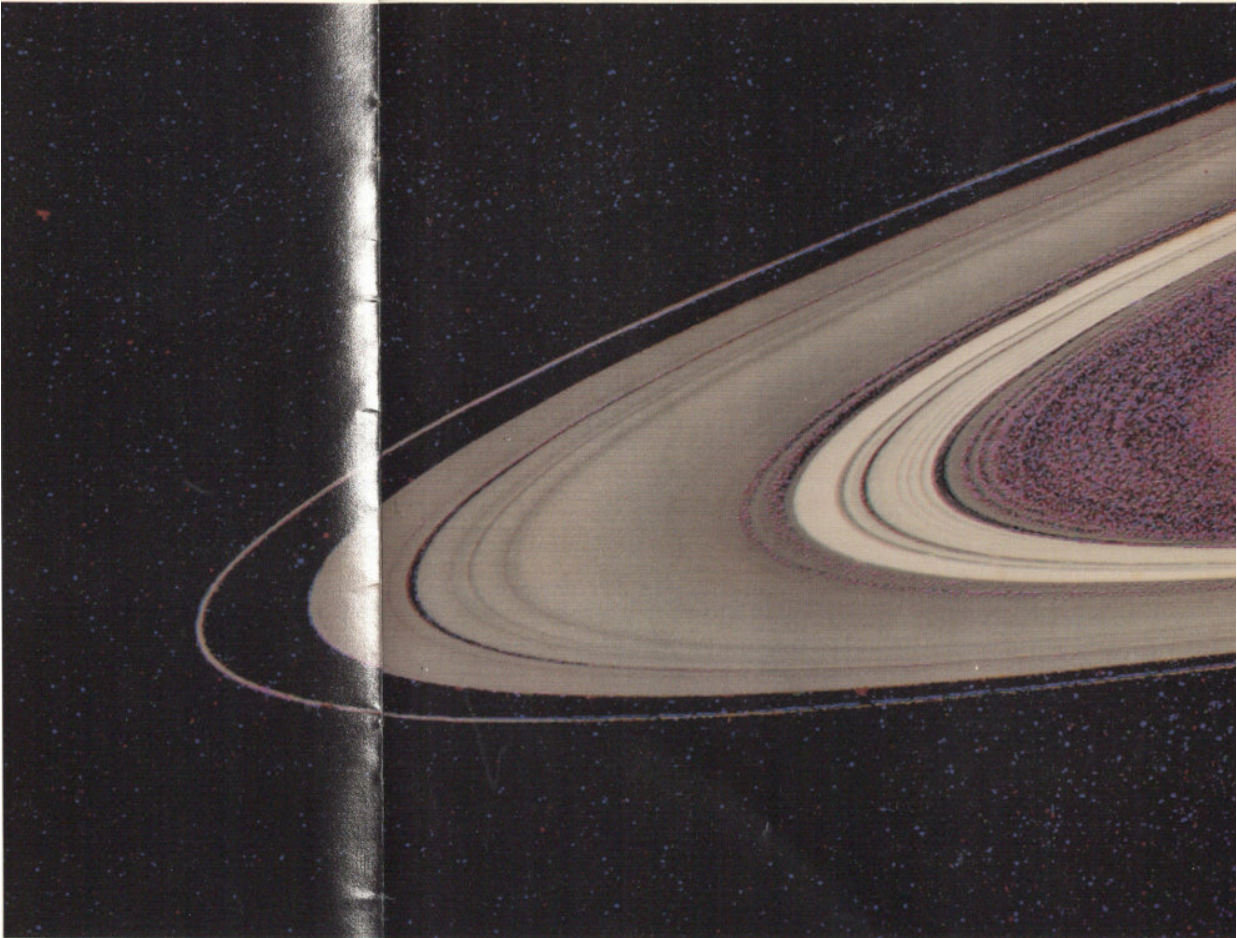


The classic features of the rings are illustrated in the diagram.

D-RING
C-RING
B-RING
"SPOKE"
CASSINI DIVISION
ENCKE DIVISION
A-RING
F-RING



11/8/80 6 million km (3.7 million mi)
The Cassini Division is filled with numerous ringlets. Discovered by Cassini in 1675, this area between the A- and B-Rings had long been thought devoid of material. The Voyager observation of well-defined rings within the Cassini Division was an unexpected discovery.



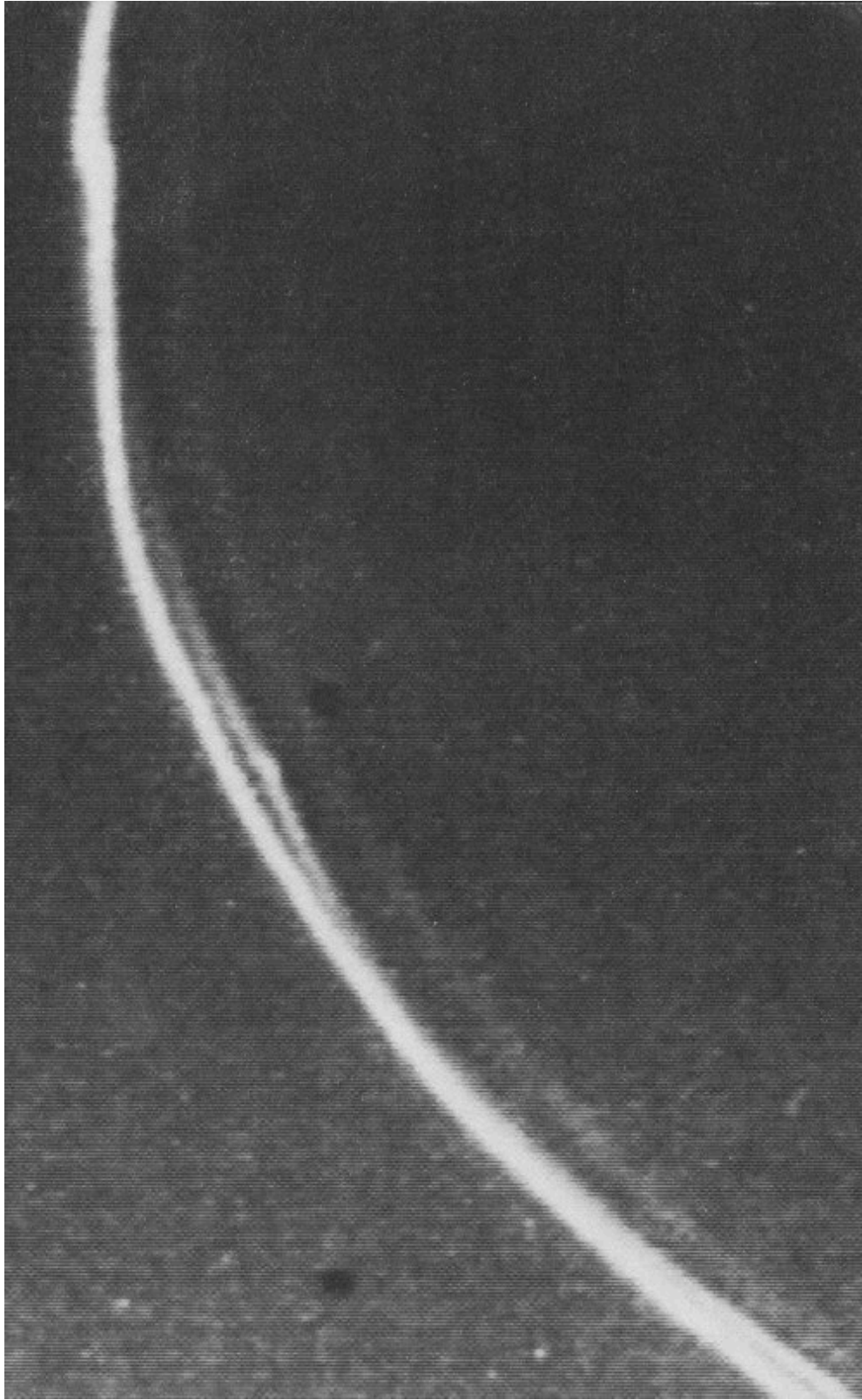
11/12/80 740,000 km (460,000 mi)

Saturn's ring system, viewed from below, appears dramatically different from its appearance on the sunlit side. This computer-processed image shows the F-Ring circling outside the A-Ring, the A-Ring with its Encke Division, the multiple ringlets in the Cassini Division, and the optically thick B-Ring, seen here in magenta hues (the coloration is an artifact of processing and is not real). The B-Ring appears dark from below the ring plane because it is dense enough to reflect most of the sunlight, causing it to appear very bright when seen from the sunward side. The opaline brightness of the Cassini Division here indicates a great deal of sunlight being scattered through this region. The Encke Division may really be empty, since it appears dark from both above and below.



11/12/80 720,000 km (450,000 mi)

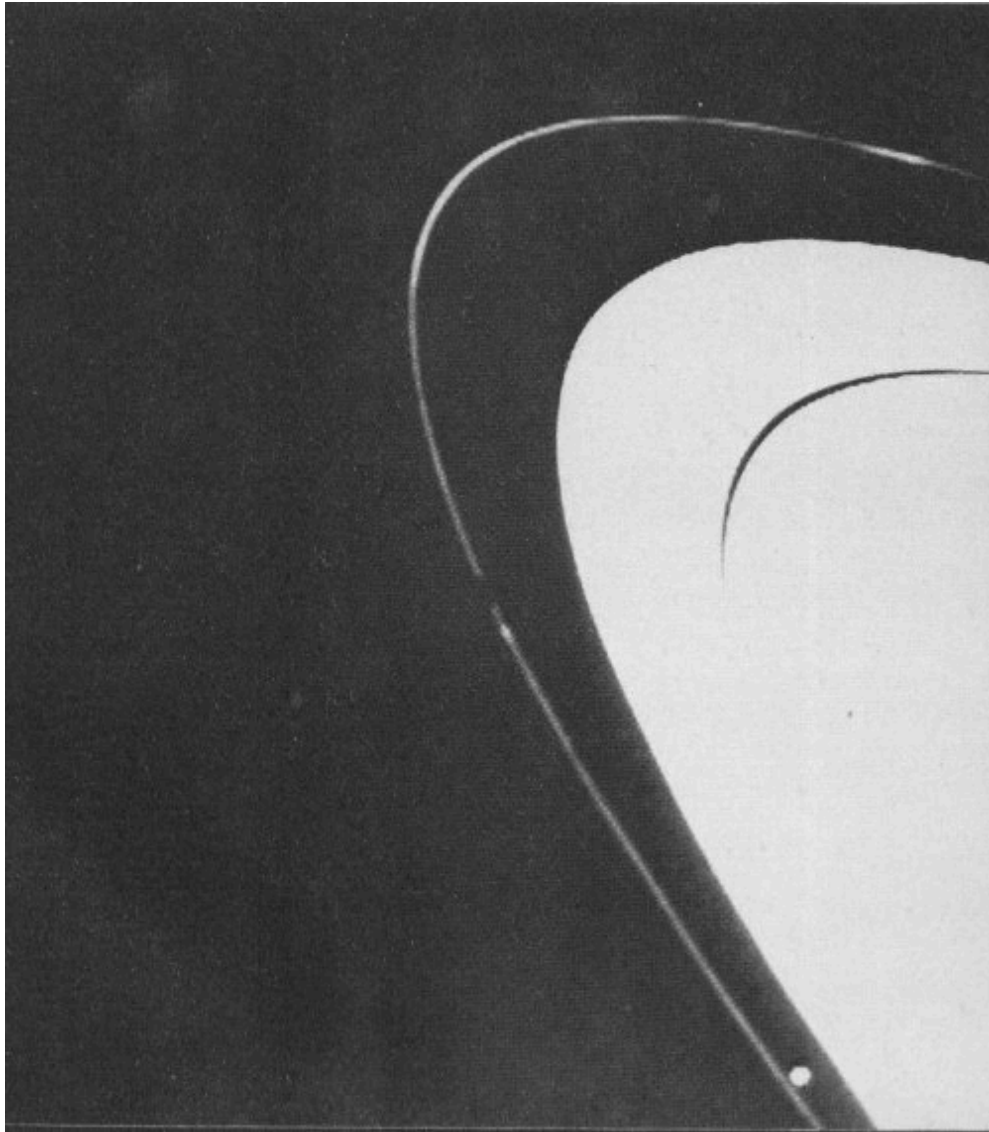
Outbound and above the ring plane, Voyager 1 gave us this view of Saturn's rings eight hours after its closest approach to the planet. The unique lighting accentuates the many hundreds of bright and dark ringlets comprising the ring system. The C-Ring (dark gray area) seems to blend into the brighter B-Ring as the concentric features radiate out from the planet. The dark spoke-like features seen in images taken during the approach to Saturn now appear as bright streaks, indicating that they may be composed of small particles.



11/12/80 750,000 km (470,000 mi)

Two narrow, braided rings in the F-Ring are evident in this view, as well as a broader, very diffuse component about 35 kilometers (20 miles) across. A totally unexpected discovery, the braided rings

trace distinctly separate orbits intertwining each other. The “knots” may be local clumps of ring material or tiny moons. It is difficult to explain this complicated structure using only the gravitational forces known to be affecting the particles of this ring. It is possible that additional, electrostatic forces may also influence these particles.

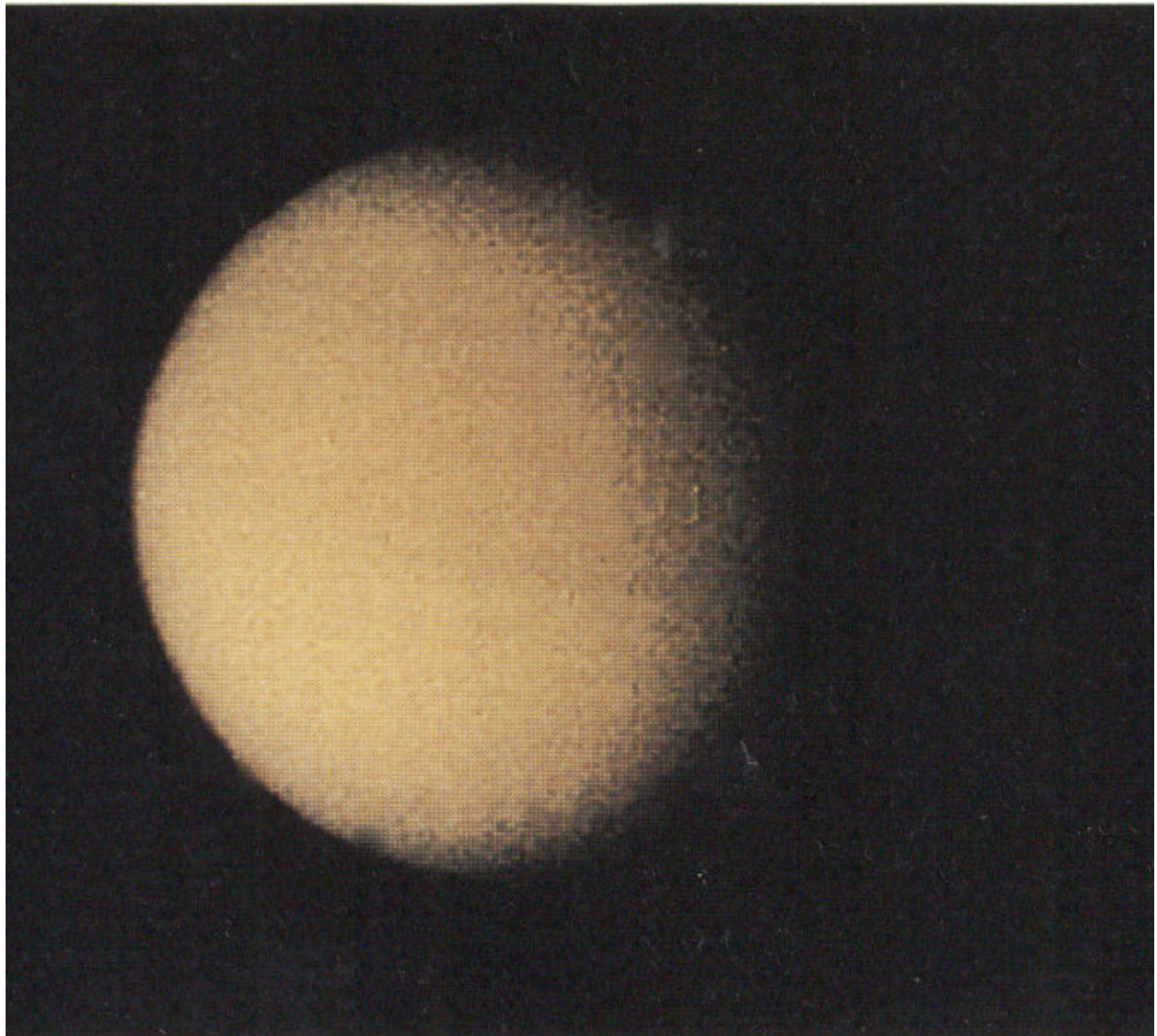


11/8/80 7 million km (4.3 million mi)

Brightness variations in the F-Ring may be due to clumping in the ring material. The features are seen at the top and again near the left edge of the ring in this image. The “gap” in the ring (left center) is not real but is the location of a reseal mark on the camera’s vidicon tube. These bright features in the F-Ring appear to move at the orbital rate of the ring particles and may be larger bodies or thicknesses in the rings. Saturn’s thirteenth and fourteenth satellites, which orbit on either side of the F-Ring, may act like “sheepdogs,” herding the F-Ring particles between them. Less than 100 kilometers (60 miles) wide, the F-Ring is located outside of the A-Ring. Satellite 14, discovered by Voyager 1, is seen just inside the F-Ring.

The Satellites

In only twelve hours, Saturn's satellites grew from names in ancient mythology into dazzling worlds with personae of their own. As Voyager 1 sailed through the Saturn system, it returned photographs of Mimas, Enceladus, Tethys, Dione, and Rhea—all part of a class of intermediate-sized icy bodies heretofore unstudied by planetary spacecraft. All but Enceladus show heavily cratered surfaces, evidence of aeons of meteorite bombardment. Enceladus hints at internal processes, as yet unidentified, which may have erased from its surface the evidence of early bombardment—but we must await Voyager 2's arrival next August to better understand this body.



11/9/80 4.5 million km (2.8 million mi)

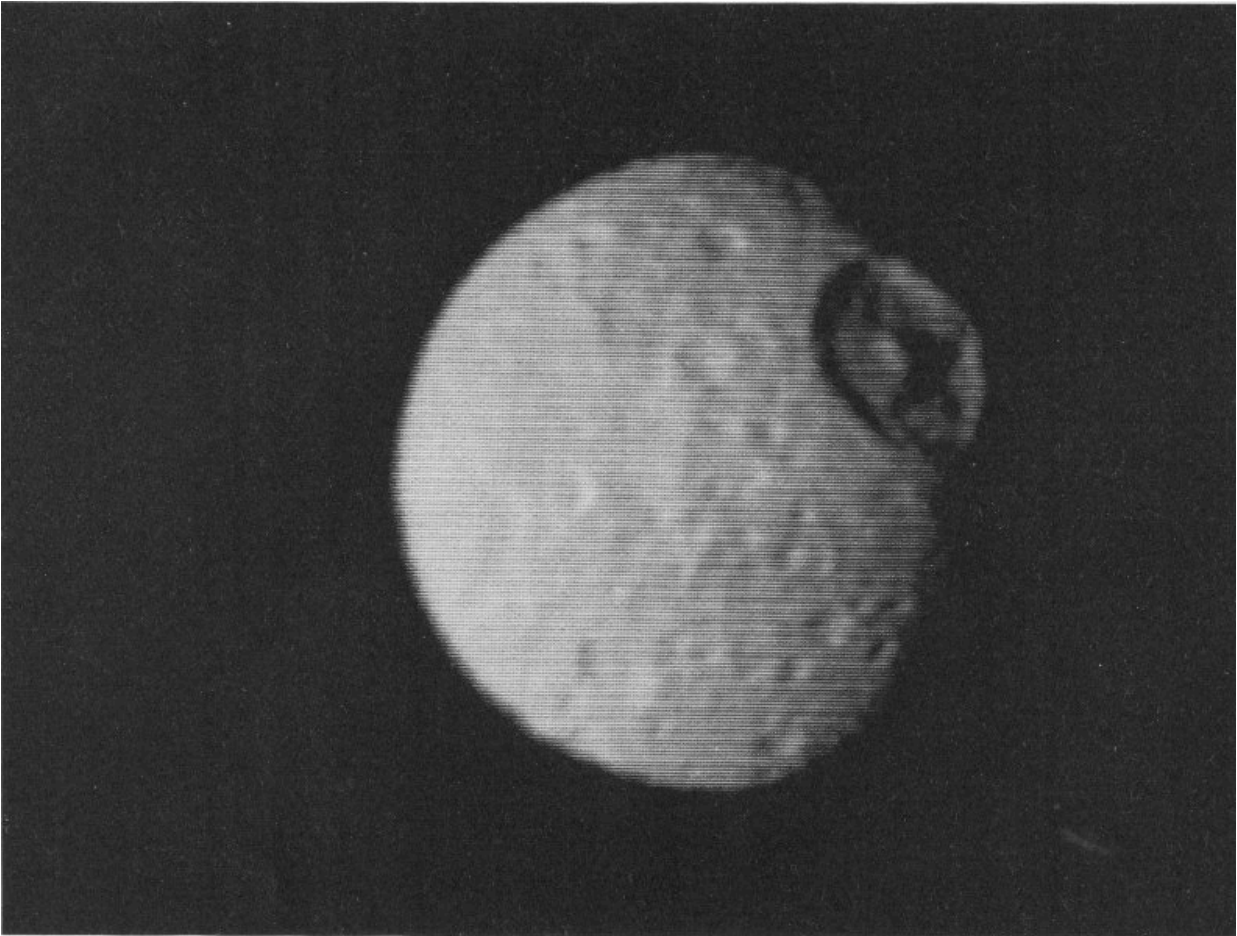
The surface of giant Titan, now dethroned from its seat as the solar system's largest satellite (Jupiter's Ganymede is larger), remains an enigma, shrouded beneath thick layers of haze.



11/12/80 22,000 km (14,000 mi)

Tiny moons—three new ones and three confirmed from previous sightings—may tell us much about ring dynamics since gravitational forces from satellites probably influence the ring structure. Two of these tiny moons are on the verge of collision in the same orbit, while several others appear to bound the A- and F-Rings. Iapetus, whose two hemispheres differ dramatically in brightness, was photographed in its orbit, almost 3.6 million kilometers (2.2 million miles) from the planet.

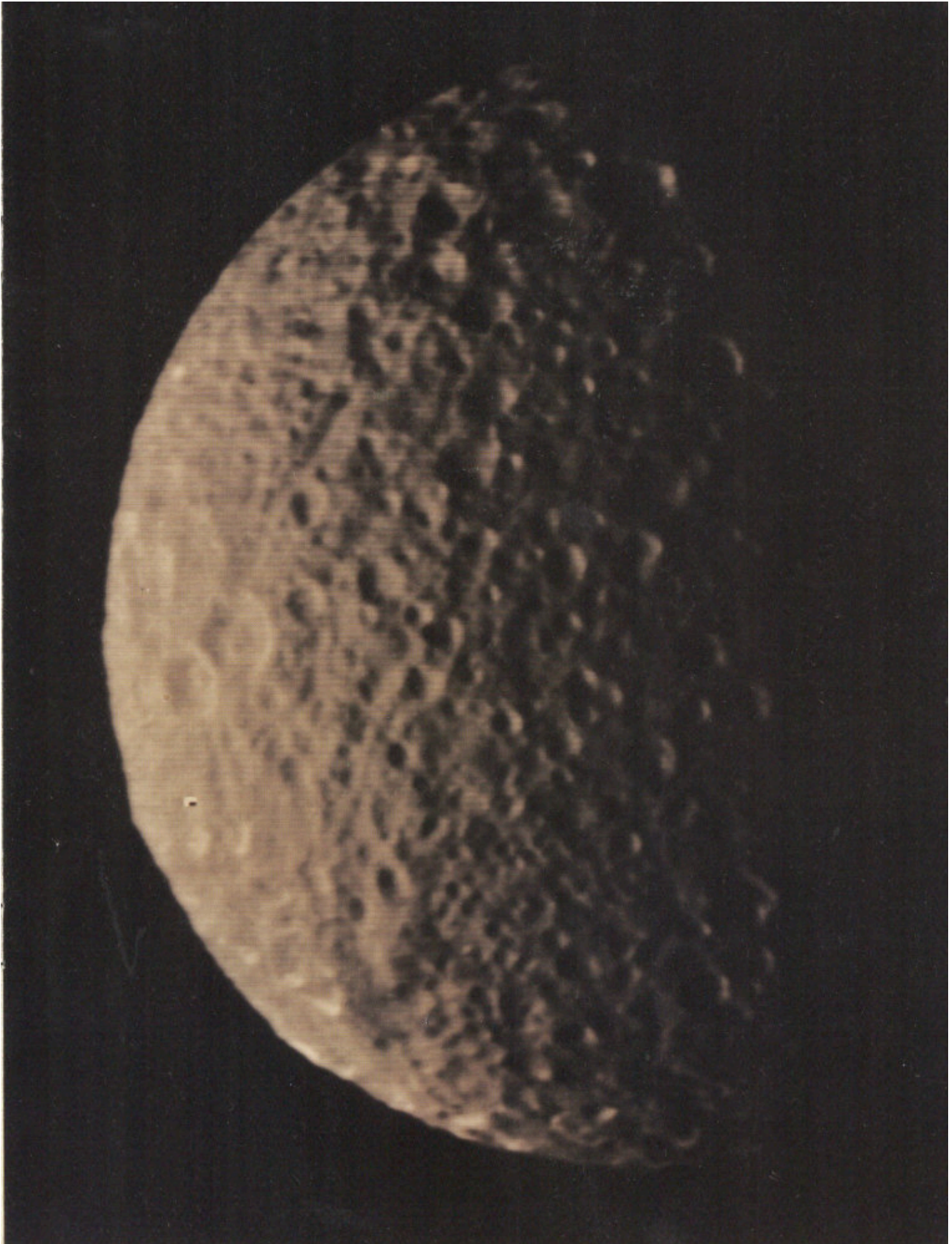
22



11/12/80 425,000 km (264,000 mi)

Mimas, Saturn's innermost large satellite, has an impact crater covering more than one quarter the diameter of the entire moon. Nowhere else in the solar system has such a disproportionately large feature been seen. In fact, it is believed that any impact larger than this would probably have shattered Mimas into two or more fragments. The crater has a raised rim and central peak, typical of large impact structures on terrestrial planets. Additional smaller craters, 15 to 45 kilometers (10 to 30 miles) in diameter, can be seen scattered across the surface, particularly along the terminator. Mimas is one of the small, low density Saturnian satellites implying that it is composed primarily of ice.

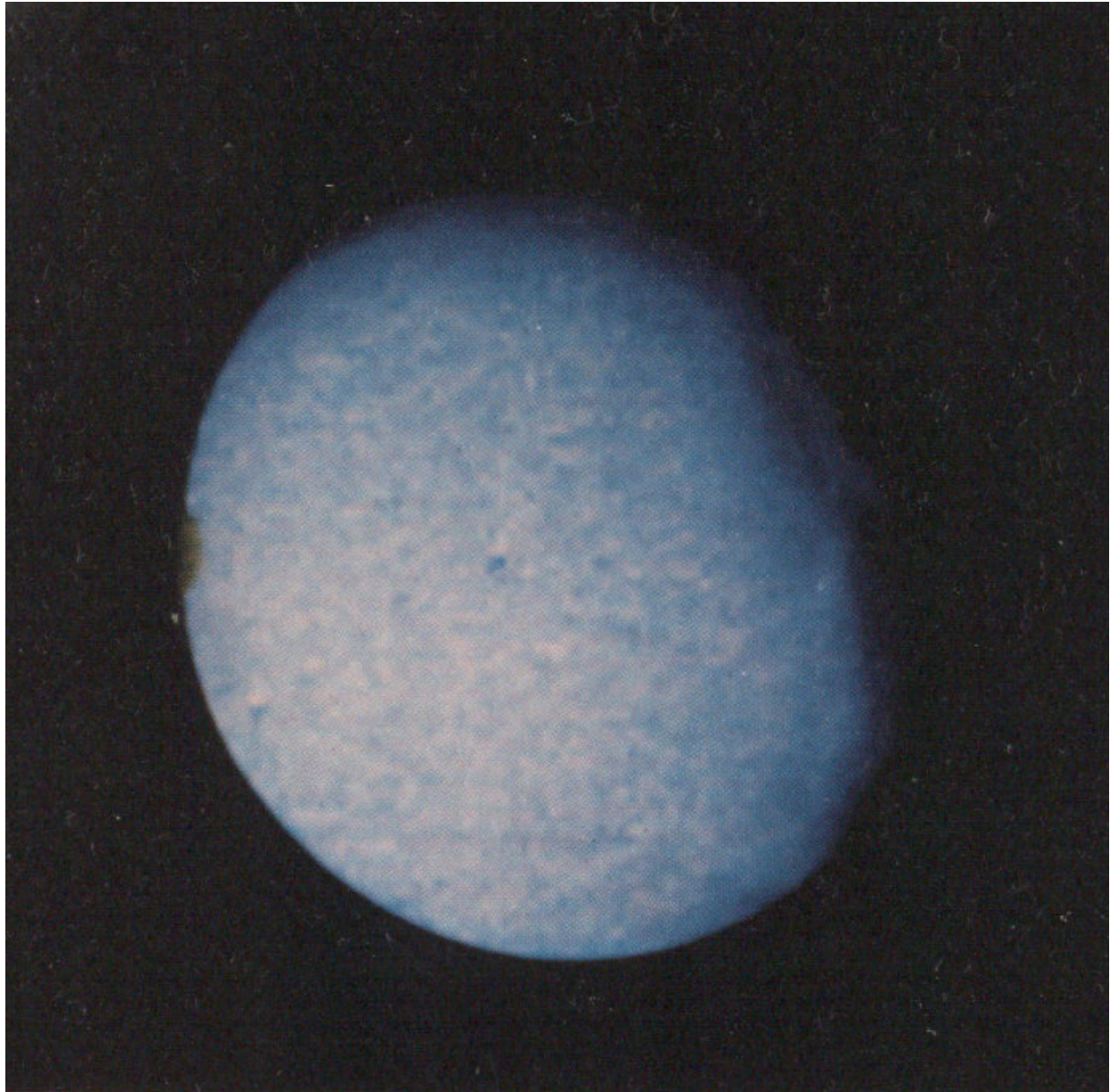
23



11/12/80 130,000 km (80,000 mi)

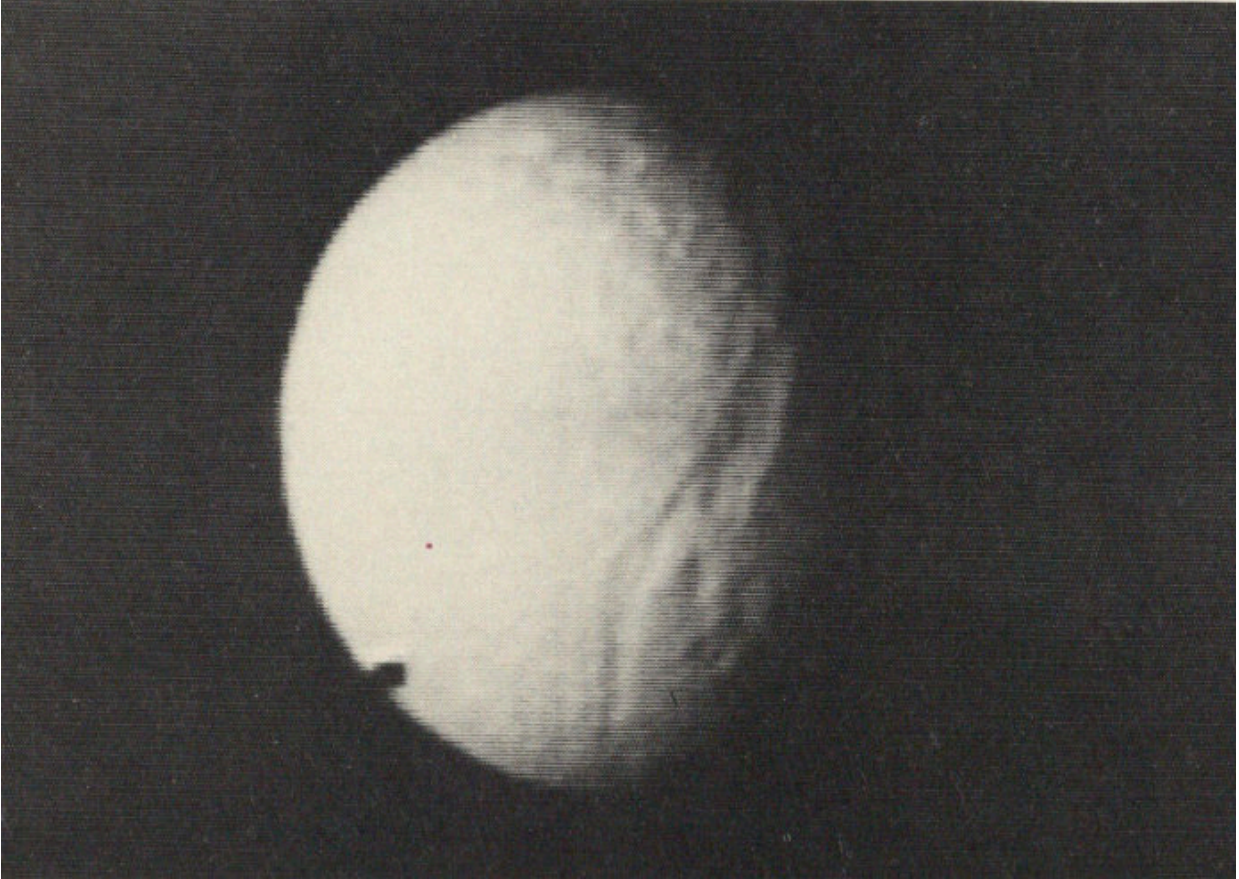
Mimas' other side shows a uniformly and heavily cratered surface—a record of the bombardment that occurred throughout the solar system in its early history 4.5 billion years ago. A long, narrow trough about 5 kilometers (3 miles) wide crosses from northeast to southwest. Mimas' surface is very reflective (about 60 percent), indicating that it consists largely of ice, which has been chipped and pulverized by aeons of meteoritic bombardment. Such a surface on a small, low mass moon would probably resemble light, powdery snow. Features as small as 3 kilometers (2 miles) across are visible.

24



11/12/80 650,000 km (400,000 mi)

Enceladus appears to be largely devoid of craters or other major surface relief, suggesting that perhaps internal processes may have erased such structures. This satellite will be seen better by Voyager 2 when it flies past Saturn in August 1981.



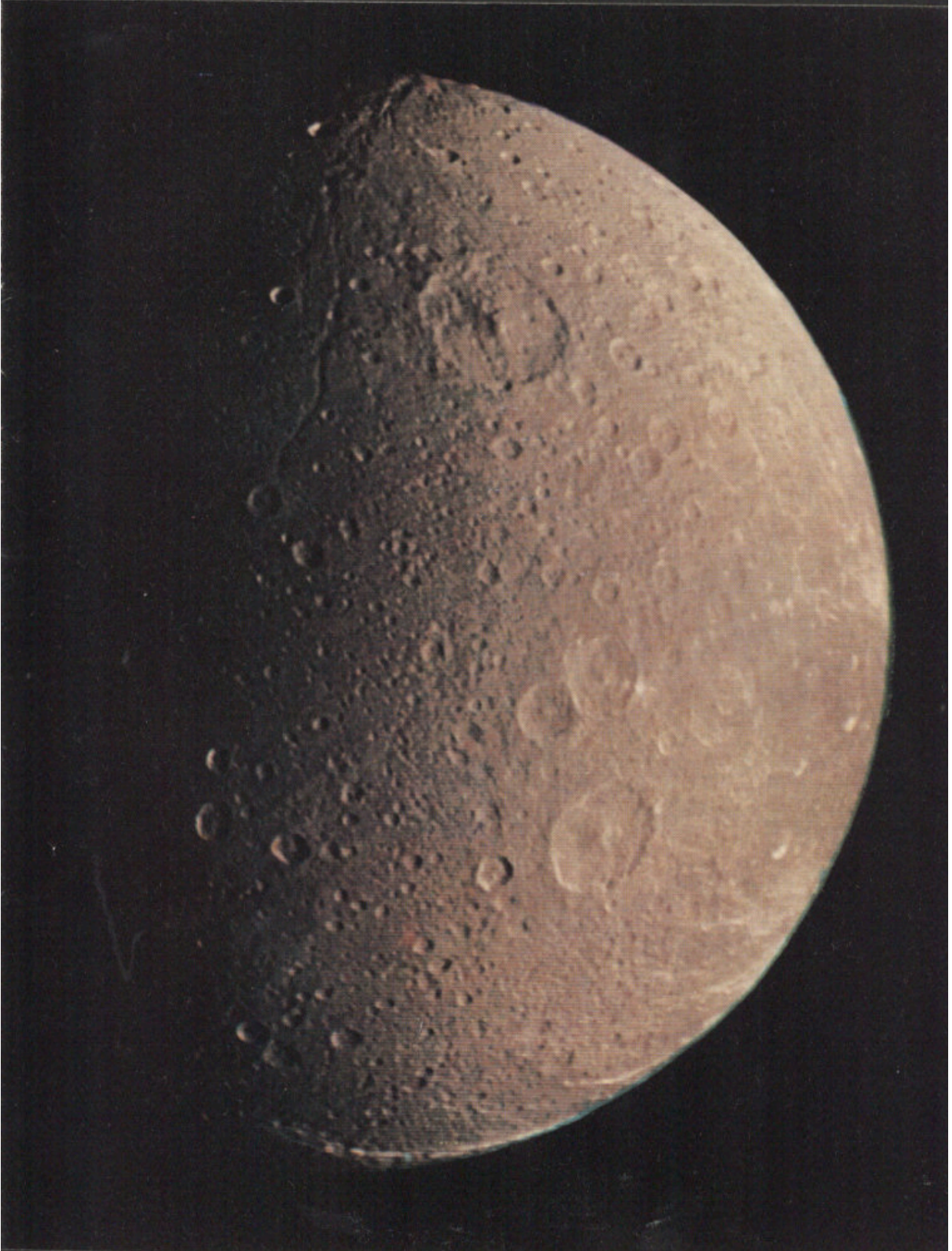
11/12/80 1.2 million km (750,000 mi)

This heavily cratered surface of Tethys faces toward Saturn and includes a large valley about 750 kilometers (500 miles) long and 60 kilometers (40 miles) wide. The craters are the result of impacts, and the valley appears to be a large fracture of unknown origin. Tethys has a diameter of 1050 kilometers (650 miles), about one-third that of Earth's Moon. The smallest features visible in this picture are about 24 kilometers (15 miles) across.



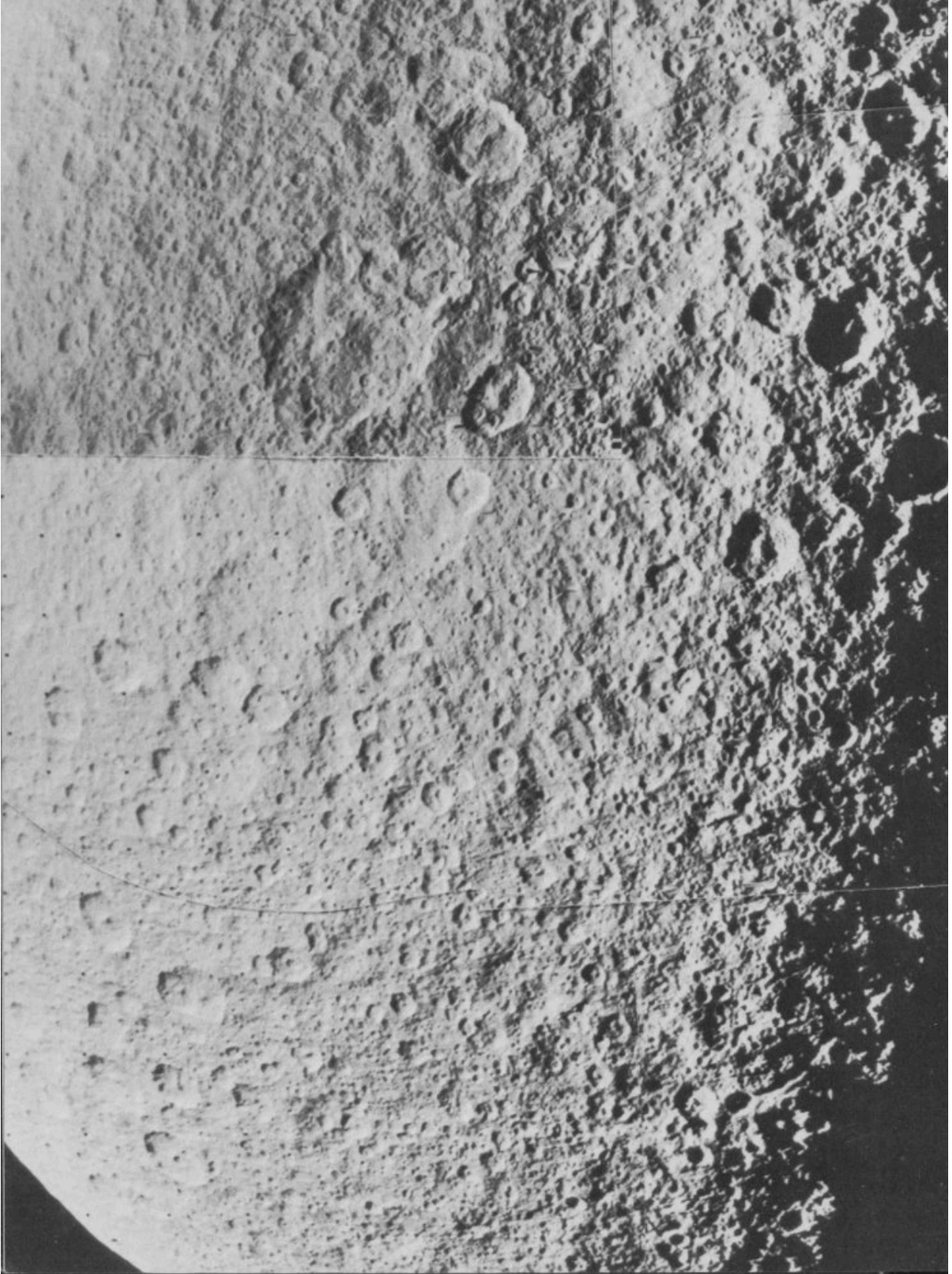
11/12/80 700,000 km (435,000 mi)

Dione reveals two distinctly different hemispheres. The photograph shows Dione's trailing side. Bright radiating patterns are probably rays of debris thrown out of impact craters; other bright areas may be topographic ridges and valleys.



11/12/80 162,000 km (101,000 mi)

Dione's other hemisphere (mosaic) also has many impact craters—the record of cosmic collisions. The largest crater is less than 100 kilometers (60 miles) in diameter and includes a well-developed central peak. Sinuous valleys (seen near each pole) are probably the result of crustal fracturing in the moon's icy crust. Dione's diameter is only 1100 kilometers (700 miles), much smaller than any of Jupiter's icy moons.



11/13/80 80,000 km (50,000 mi)

Craters stand shoulder-to-shoulder on the surface of Saturn's satellite Rhea, seen in this mosaic of the highest-resolution pictures of the north polar region. Rhea is 1500 kilometers (950 miles) in diameter and is the most heavily cratered Saturn moon. The largest crater, made by the impact of cosmic debris, is about 300 kilometers (190 miles) in diameter.

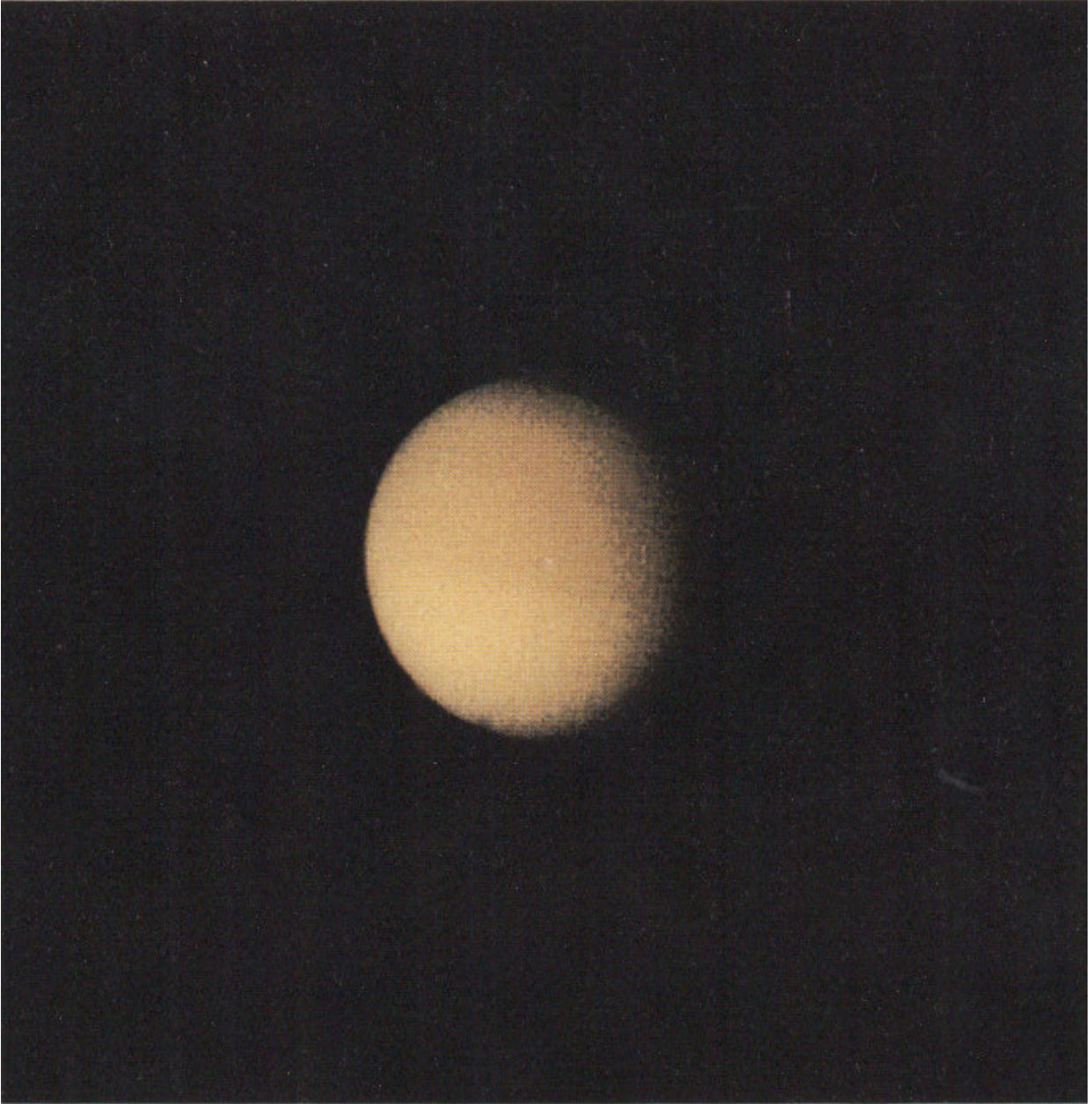
27



11/12/80 128,000 km (79,500 mi)

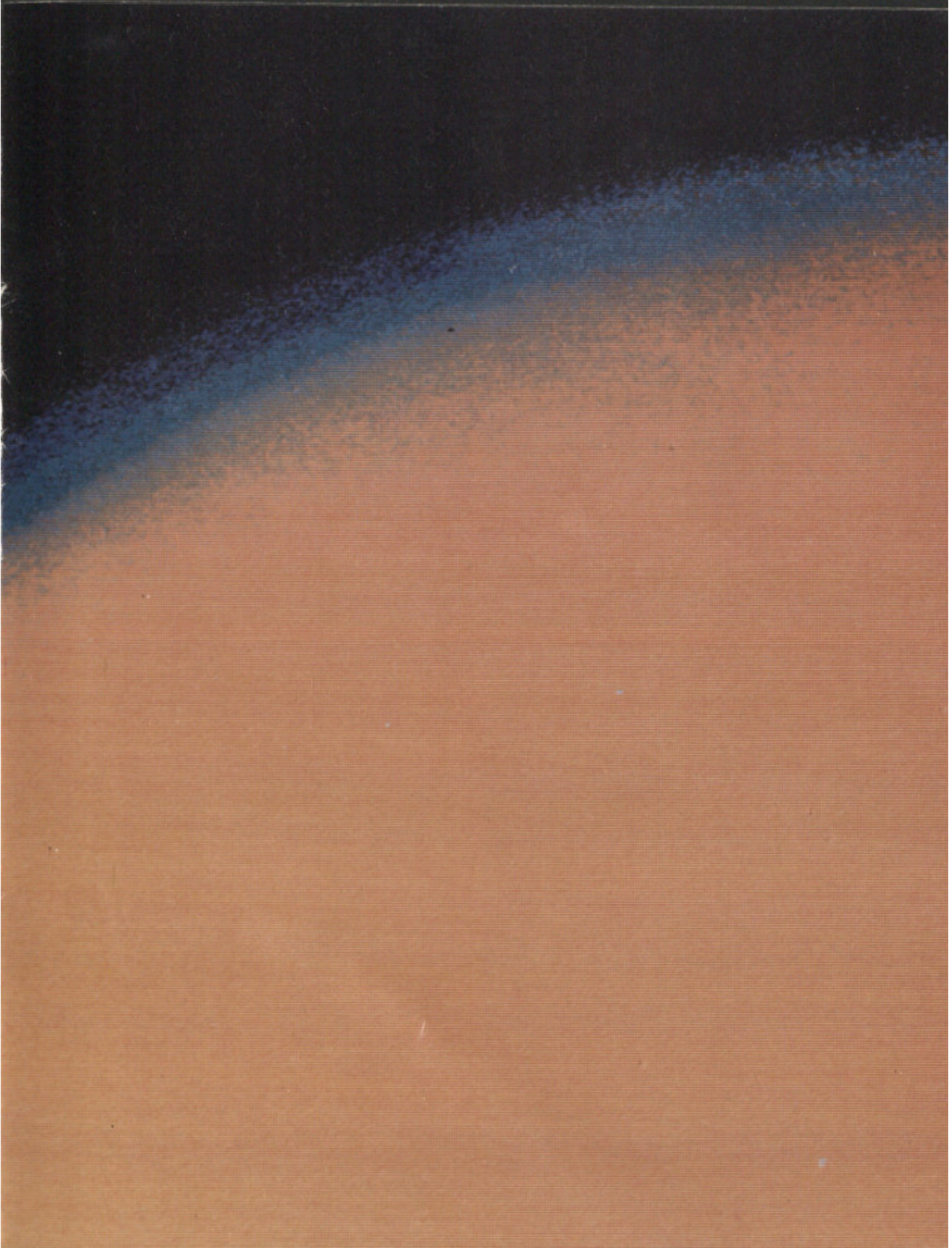
Impact craters on the ancient surface of Rhea closely resemble those on Mercury and Earth's Moon. Many of the craters have central peaks formed by rebound of the floor during the explosive formation of the crater. Some craters are old and degraded by later impacts. Many have sharp rims and appear relatively fresh, while others are very shallow and have subdued rims, indicative of their antiquity. White areas on the edges of several of the craters are probably fresh ice exposed on steep slopes or possibly deposited by volatiles leaking from fractured regions. Surface features as small as 2.5 kilometers (1.5 miles) in diameter are visible.

28



11/9/80 4.5 million km (2.8 million mi)

Titan is a large, bizarre satellite. It is larger (almost 5120 kilometers or 3180 miles in diameter) than the planet Mercury and possesses a dense atmosphere of unique composition. Voyager 1's cameras show Titan's surface to be totally obscured by a thick layer of atmospheric haze. In the full-disk photograph, only two features are visible: a faint boundary between the southern and darker northern hemispheres and a dark "hood" overlying Titan's north polar region.



11/12/80 435,000 km (270,000 mi)

This hood and greater detail in the haze layers are shown in the higher resolution photograph.



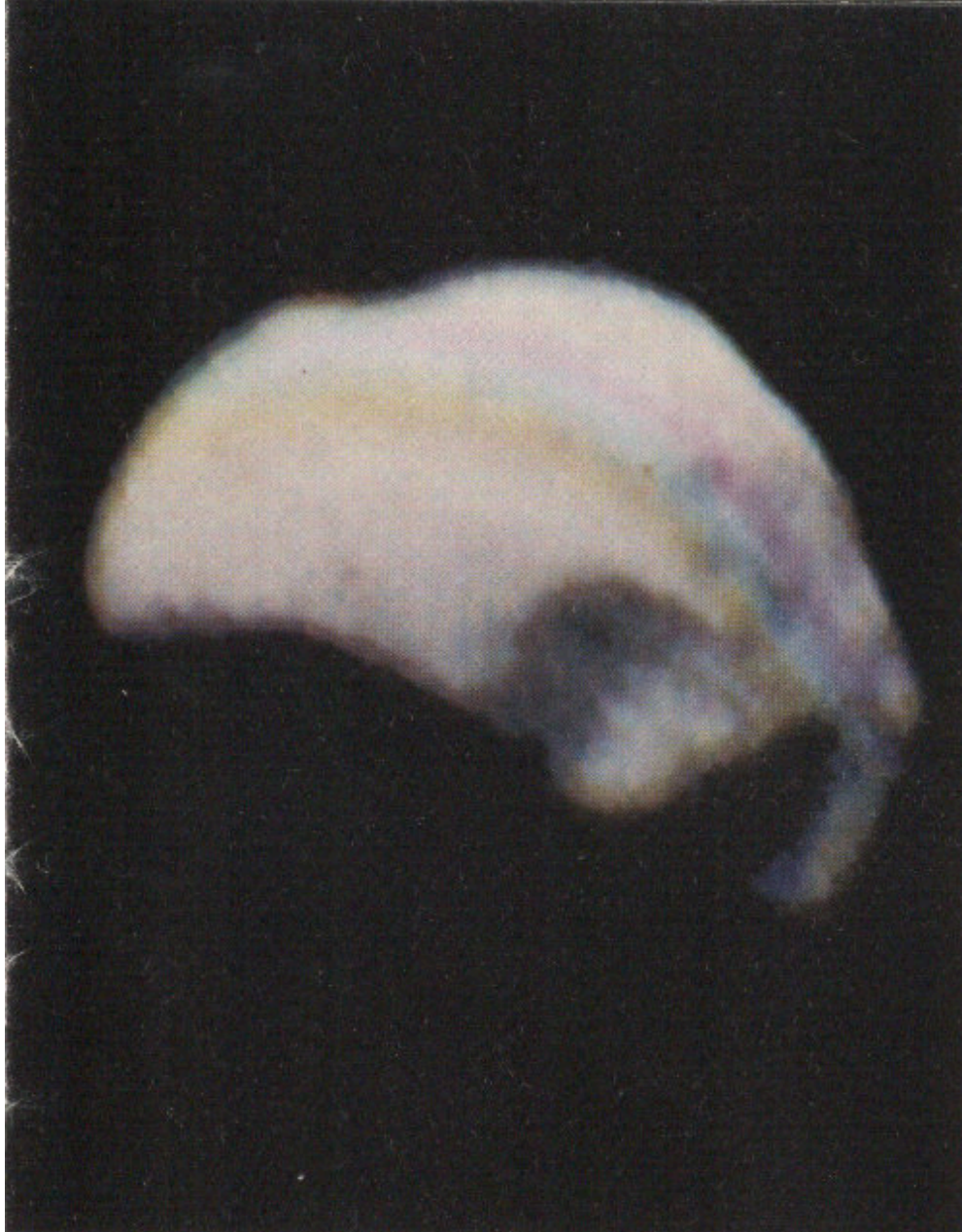
11/10/80 4.6 million km (2.8 million mi)

Little detail can be seen in this distant view of Hyperion, the satellite which orbits just beyond Titan. Voyager 2 will observe Hyperion at a closer range.



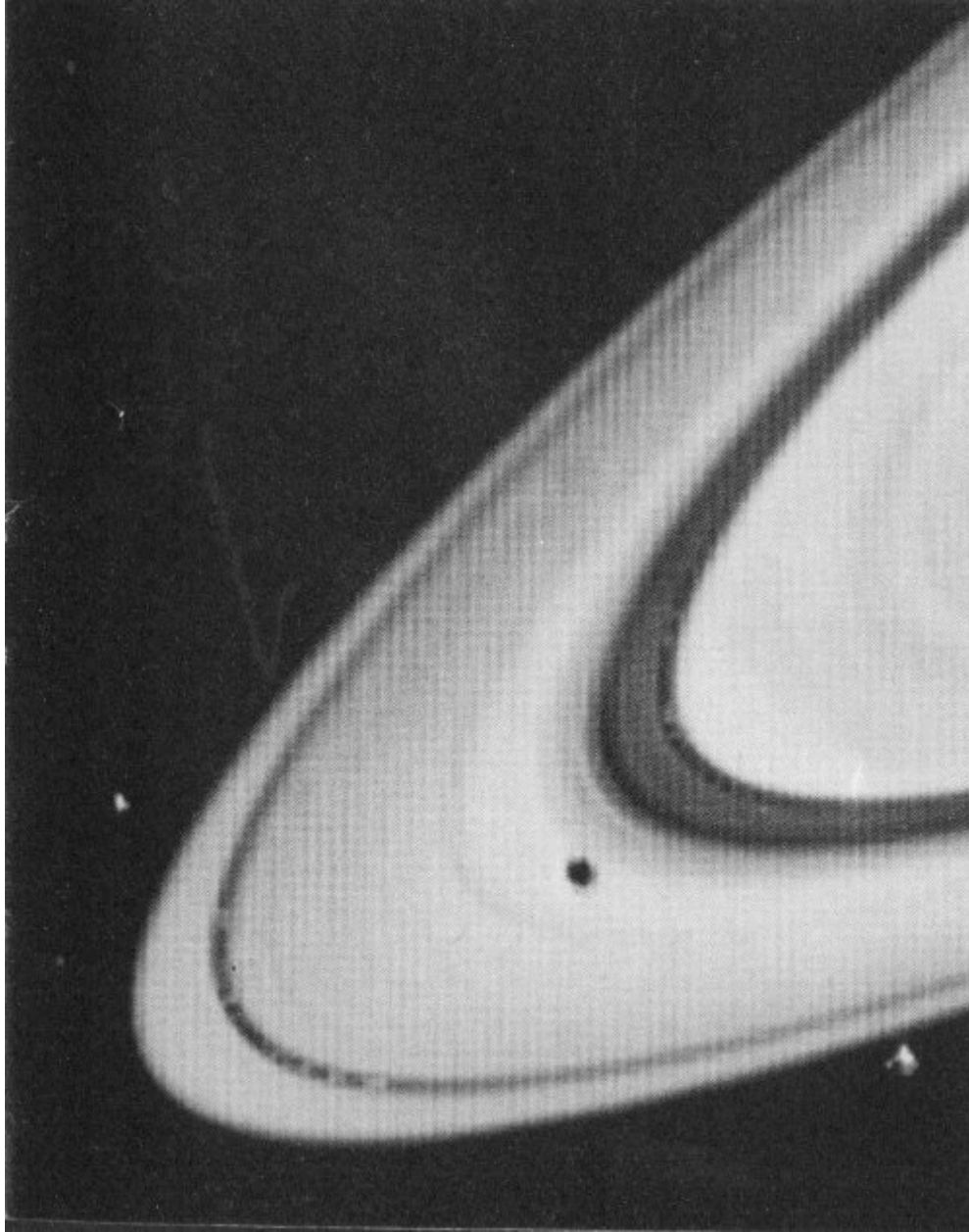
11/12/80 3.2 million km (1.9 million mi)

Saturn's satellite Iapetus displays a large, circular feature about 200 kilometers (120 miles) across with a dark spot in its center. The circular feature is probably a large impact structure outlined by dark material, possibly thrown out by the impact. The satellite's leading hemisphere is to the left, and the trailing hemisphere, which is four to five times brighter, is to the right. Iapetus' diameter is 1450 kilometers (900 miles).



11/12/80 177,000 km (110,000 mi)

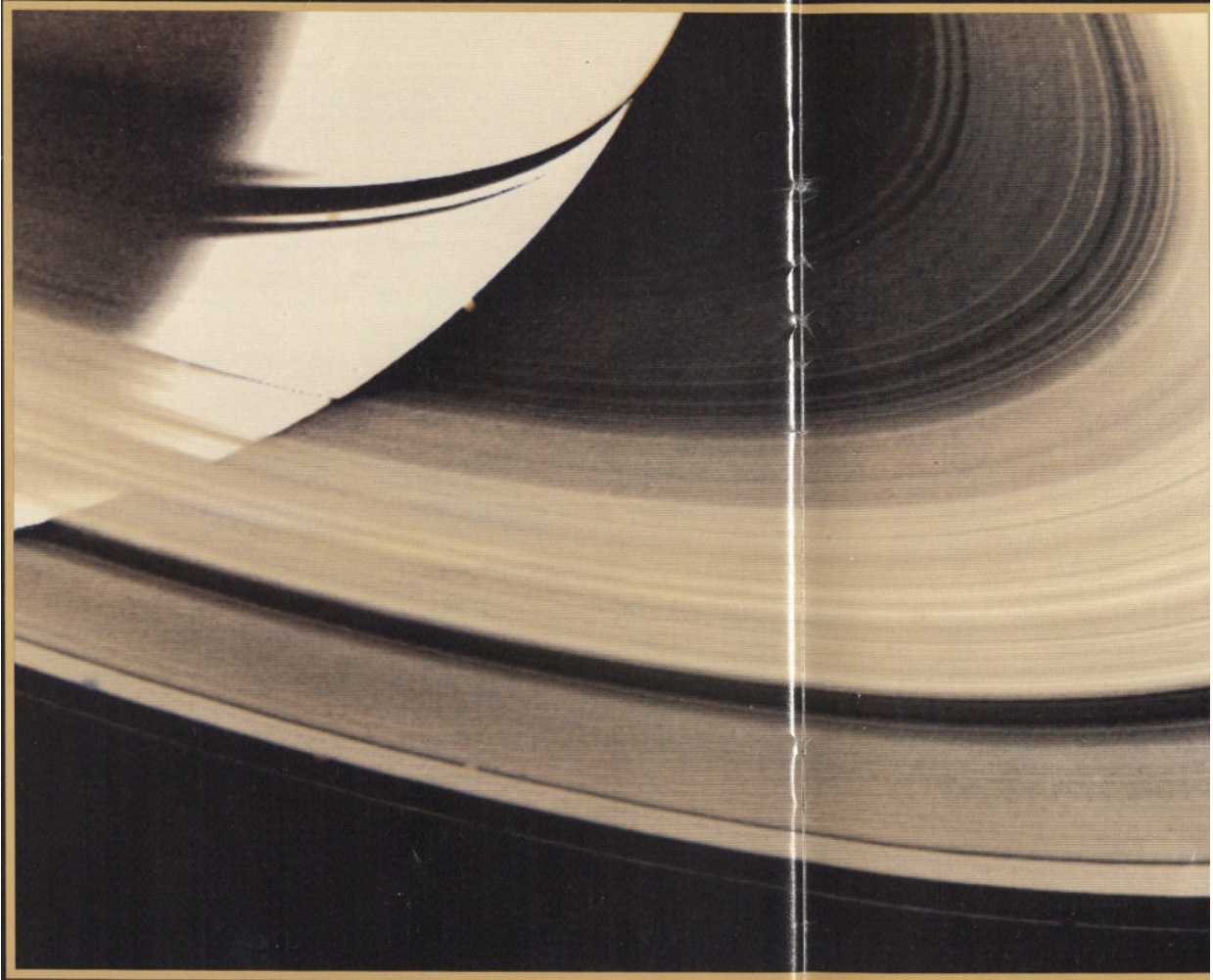
Two satellites (Saturn's tenth and eleventh) revolve in nearly identical orbits 151,000 kilometers (94,000 miles) from Saturn's center. The satellites are each 100 to 200 kilometers in diameter, larger than the distance separating their orbits, and they are currently approaching one another at a rate which promises collision in about two years. Such a collision, however, will probably be averted by orbital changes induced by the satellites' mutual gravitational interactions as they near one another. The trailing co-orbital satellite, seen in this photograph, has a very irregular outline (the Sun is shining from the left). This color composite was produced from three exposures taken over a period of more than six minutes. During this period, a thin shadow, cast by a previously unknown ring, moved across the satellite causing the "rainbow" pattern shown here.



10/25/80 25 million km (16 million mi)

Two smaller satellites—Saturn's thirteenth and fourteenth moons—were discovered on October 25, 1980, in images taken to study the dark “spokes” within Saturn's B-Ring. The smaller, inner satellite has a diameter of about 500 kilometers (300 miles) and is visible just outside the A-Ring, near the bottom of the picture. It travels in an orbit between the A-Ring and the F-Ring (not visible in this photograph). The second satellite, seen to the left, travels just outside the F-Ring and is about 600 kilometers (400 miles) in diameter. Scientists believe the dimensions of the narrow F-Ring may be determined by these two satellites, which orbit on either edge of the ring.

A Glimpse Back



11/13/80 1.5 million km (930,000 mi)

Looking back at the Saturn system as it soared upward and outward, Voyager 1 continued its observations for nearly five weeks after closest Saturn approach. The spacecraft photographed the planet's sunlit crescent, the ring shadows falling on the planet, and Saturn's dark hemisphere illuminated by "ringshine." It searched for lightning and auroras on the planet's dark side and looked for "sun dogs" resulting from ammonia crystals in the atmosphere. It continued temperature and composition measurements and searched for new satellites out to the orbit of Mimas. It measured the flow of plasma in Saturn's magnetosphere and now, its journey far from over, Voyager 1 proceeds toward the outer boundary of our solar system, as it seeks to probe the space among the stars of our galaxy, the Milky Way.



11/16/80 5.3 million km (3.3 million mi)
Departing Saturn, Voyager 1 photographed the planet from a unique perspective, clearly showing Saturn's shadow on the rings.



11/12/80 250,000 km (150,000 mi)

During a 40-minute period on the day of encounter, the spacecraft was itself in the planet's shadow. At this time, the wide-angle camera acquired a photograph of this shadow line, revealing ring material in a region very close to the planet, where no material had been previously observed. This inner ring, the

D-Ring, is roughly 6000 kilometers (4000 miles) wide and extends to within about 6000 kilometers of Saturn's cloudtops.

The Voyager Mission

Only once every 175 years are the outer planets aligned in their orbits so that we can take advantage of gravity-assist trajectories to achieve encounters with Jupiter, Saturn, Uranus, and Neptune on one mission. The gravity-assist technique uses one planet's gravity field and motion through space to alter the spacecraft's flight path and propel it outward toward the next planet. Voyager 1's trajectory, which was selected to best view Titan, has now propelled the spacecraft out of the ecliptic plane, while Voyager 2's path will remain in this plane to provide future encounters with Uranus and possibly with Neptune.

MISSION OBJECTIVES

The Voyager Project was approved in June 1972 and had as its mission objectives:

- ★ Exploration of the Jupiter and Saturn planetary systems, including their atmospheres, rings, satellites, and magnetospheres
- ★ Comparative analyses of the two systems
- ★ Investigation of the interplanetary medium between Earth and Saturn

A fourth objective, added in 1976, was to preserve the possibility of extending the mission to include an investigation of the planet Uranus and the interstellar medium.

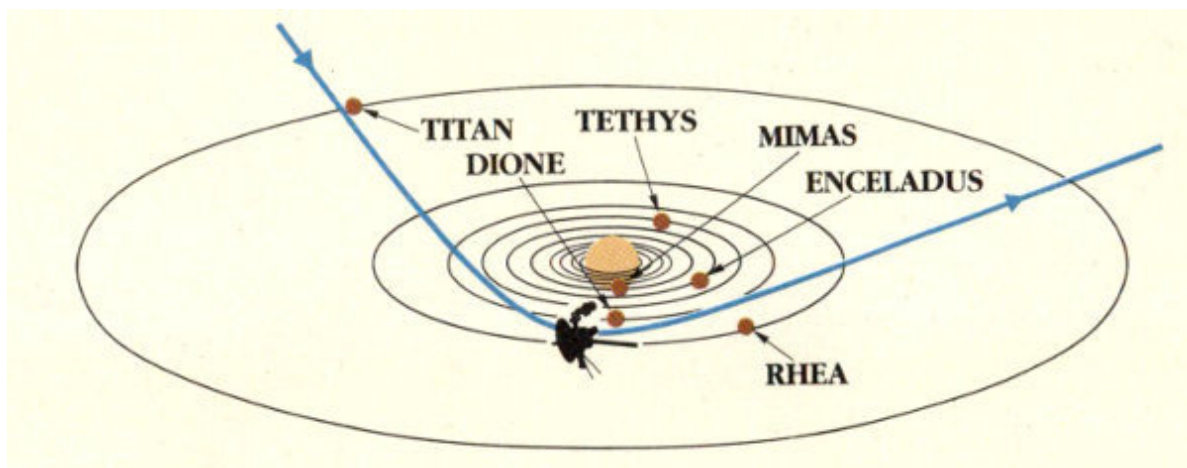
With the completion of Voyager 1's Saturn flyby, it is now clear that these objectives will be achieved.

SPACECRAFT CHARACTERISTICS

Two identical spacecraft were developed for the 1977 launch opportunity. These marvelous machines were cleverly designed to survive the rigors of long voyages in outer space and to deliver high-quality scientific information required for detailed understanding of planetary systems. The spacecraft are both complex—automatically responding to their Earth-bound monitors that remotely control them via radio commands—and highly autonomous—capable of caring for themselves in many areas through a system of sensors, computers, and spare equipment. Each spacecraft functions on about 400 watts of electrical power which is provided by nuclear generators. Broadcasts of data across a billion miles to Earth are accomplished with a spacecraft transmitter power of only about 25 watts, the amount of energy required by a small household light bulb.

Voyager's scientific payload was carefully chosen to observe Saturn over a wide range of wave-lengths and to measure magnetic fields, charged particles, and plasma waves.

SATURN ENCOUNTER

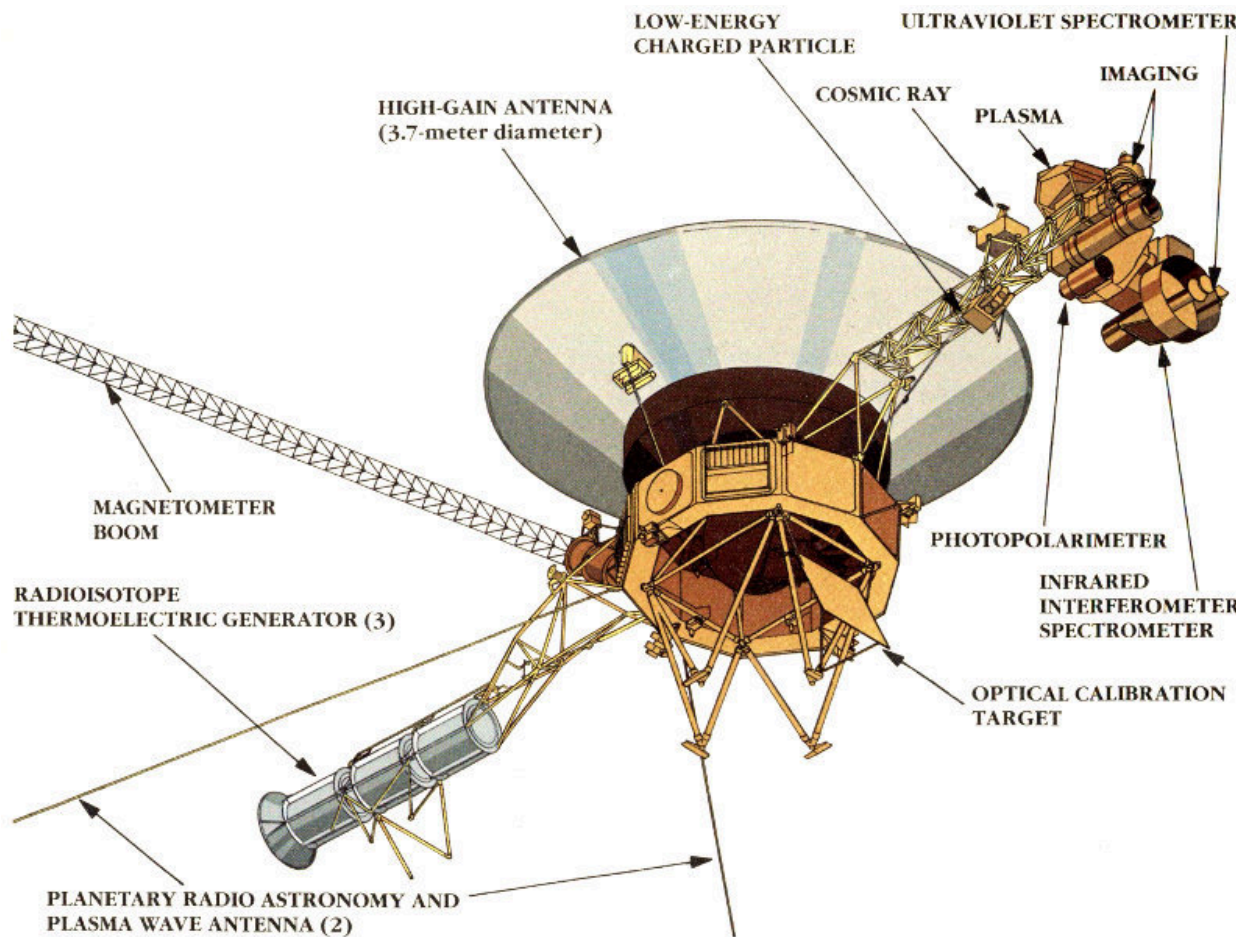


Voyager 1 approached within 124,000 kilometers (77,000 miles) of Saturn's cloudtops. Six of the satellites that were photographed are shown in their approximate positions at closest approach by the spacecraft.

TITAN
DIONE

TETHYS
MIMAS
ENCELADUS
RHEA

Voyager 1's Saturn encounter period began on August 22, 1980, at a range of 109 million kilometers (68 million miles) from the planet. Even at this great distance, Voyager's images were better than any from Earth-based telescopes. During the long encounter period, which extended through December 19, 1980, continuous observations of Saturn's realm were carried out by Voyager's instruments. Voyager 1's flight path through the Saturn system demanded navigation of the highest precision to meet three critical targets: (1) a close 4000-kilometer (2300-mile) flyby and occultation at Titan, (2) a precise, three-minute time period when the spacecraft was emerging from occultation at the same time Earth was in a position to receive the spacecraft signals passing through the gap between Saturn and its rings, and (3) a flight path through the E-Ring at Dione's orbit to assure safe passage through a zone clear of potentially dangerous material. To assure these targets were achieved, small trajectory trim maneuvers were executed on October 11, 1980, and again on November 6, 1980, as Voyager 1



Voyager spacecraft and scientific instruments.

HIGH-GAIN ANTENNA (3.7-meter diameter)
LOW-ENERGY CHARGED PARTICLE
COSMIC RAY
PLASMA
IMAGING
ULTRAVIOLET SPECTROMETER
INFRARED INTERFEROMETER SPECTROMETER
PHOTOPOLARIMETER
OPTICAL CALIBRATION TARGET
PLANETARY RADIO ASTRONOMY AND PLASMA WAVE ANTENNA (2)
RADIOISOTOPE THERMOELECTRIC GENERATOR (3)
MAGNETOMETER BOOM

By October 24, 1980, when Voyager 1 was about 30 million kilometers (19 million miles) from Saturn, the spacecraft's narrow-angle camera could no longer capture the planet in a single picture. Thus, a period of multiple images or mosaics began. By November 2, 1980, even four-picture mosaics

could no longer cover the rapidly growing scene. Voyager 1's pace of operations reached an exciting peak during the near-encounter phase from November 11 through November 13, 1980. While still about 1.6 million kilometers (1 million miles) from closest approach to Saturn, Voyager 1 encountered Titan on November 11, 1980, and then dipped below the ring plane as it accelerated rapidly toward Saturn. On November 12, 1980, Voyager 1 came within 124,000 kilometers (77,000 miles) of the cloudtops of Saturn's southern hemisphere, where Saturn's gravity altered the spacecraft's course, hurtling the spacecraft upward past the ring plane. Close observation of Saturn's other major satellites and its rings were made during this passage.

From Earth to Saturn, Voyager 1 has traveled in the ecliptic plane, the plane in which the major planets orbit. Now, having completed its final planetary flyby, Voyager 1 is rising above this plane on a trajectory that will eventually carry it above and out of the solar system, probably before the end of this century. As it proceeds, the spacecraft will return information about the solar wind and magnetic fields in the far, unexplored reaches of our solar system and will observe cosmic rays emitted from the distant stars among which Voyager will ultimately cruise.

Scientific Highlights

Some of the most important information gathered by Voyager 1 on the Saturn system is presented pictorially in this publication and is supplemented here with brief summaries of the major discoveries, observations, and theories.

SATURN

Saturn's atmosphere appears similar to Jupiter's, with alternating dark belts and bright zones, circulating storm regions, and other dark and light cloud markings. Saturn's belt and zone system extends to higher latitudes than those on Jupiter, and all of the features are muted by a thick atmospheric haze, perhaps 70 kilometers (40 miles) deep.

Wind speeds up to 1500 kilometers per hour (900 miles per hour) occur at the equator—four to five times faster than any Jovian winds. Temperatures near the cloudtops range from 86 to 92 kelvins (-305° to -294° Fahrenheit)—nearly 60 degrees colder than at Jupiter. Saturn still radiates about 2.8 times as much heat as it receives from the Sun. The coolest temperatures are found at the center of the equatorial zone.

Auroral emissions have been seen near Saturn's poles, and auroral-type emissions have been seen in ultraviolet light near the illuminated limb of the planet.

Lightning bolts have not been seen on Saturn, but radio emissions typical of lightning discharges have been recorded. The source of these discharges is believed to be the rings rather than Saturn's atmosphere.

RINGS

Hundreds of tiny ringlets—a few of them elliptical rather than circular—comprise the classic A-, B-, and C-Rings, once thought to be uniform disks of material. The F-Ring, which was first sighted by Pioneer 11 in 1979, was observed to be three separate, intertwined ringlets.

The existence of a D-Ring between the C-Ring and the planet has been confirmed by observations during Voyager 1's passage through Saturn's shadow. The tenuous E-Ring, previously observed from Earth only when Saturn's rings could be viewed edge-on (every 15 years), has also been observed during shadow passage. At least one other ring has been found between the E- and F-Rings in Voyager images.

Long, radial, spoke-like features in the B-Ring were dark when viewed upon approach and bright when observed after encounter when the spacecraft looked back toward the planet and the Sun.

NEW SATELLITES

Voyager 1 photographed six tiny moons, some that had never been seen before. Satellites 10 and 11, dubbed the “co-orbitals,” share an orbit 91,000 kilometers (57,000 miles) above Saturn's cloudtops. The leading satellite has a diameter of about 160 kilometers (100 miles), while the trailing satellite has an irregular shape, approximately 105 by 65 kilometers (65 by 40 miles).

Little is known about satellites 12, 13, 14, and 15 aside from their orbits and periods. Satellite 12 orbits at the same distance from Saturn as Dione, at a point about 60 degrees ahead of Dione. Satellites 13 and 14, outside and inside the F-Ring (respectively), appear to “herd” this thin ring between them. Satellite 15 appears to limit the outer edge of the A-Ring in a similar manner.

INNER SATELLITES

Mimas, Enceladus, Tethys, Dione, and Rhea represent a body size not previously explored by spacecraft. They are larger than Jupiter's Amalthea

and Mars' Phobos and Deimos, yet smaller than Mercury, our Moon, or Jupiter's large satellites. Their diameters range from 390 kilometers (240 miles) for Mimas to 1530 kilometers (950 miles) for Rhea, and they are probably composed primarily of water ice.

With the exception of Enceladus, all of these moons have heavily cratered surfaces, looking much like the Moon and Mercury. Mimas displays an impact crater whose diameter is one-fourth that of the satellite—such an impact must have nearly shattered the icy satellite. Tethys has a valley 70 kilometers (40 miles) wide that stretches 800 kilometers (500 miles) across the satellite, an apparent crustal fracture resulting from seismic activity. Several sinuous valleys, some of which appear to branch, are visible on Dione's surface. Both Dione and Rhea have bright, wispy streaks on their already highly reflective surfaces, perhaps caused by ice thrown out of craters by meteorite impacts.

39

Of the five inner moons, Enceladus appears the smoothest, but we will have to wait for Voyager 2 to photograph the satellite at greater resolution in 1981. Since the maximum intensity of the E-Ring occurs near Enceladus' orbit, Enceladus may be a source of E-Ring particles.

TITAN

Titan is now known to be smaller than Jupiter's Ganymede. Its diameter is less than 5120 kilometers (3180 miles), which implies a density twice that of water ice. A dense, hazy atmosphere at least 400 kilometers (250 miles) thick obscures the surface. Voyager 1 determined that Titan has a nitrogen-rich atmosphere (as does Earth), but with concentrations of hydrocarbons such as methane (natural gas), ethane, acetylene, ethylene, and deadly hydrogen cyanide. The haze layers merge into a darkened hood over the north pole. At the poles, liquid nitrogen lakes may form. The surface temperature is probably near 100 kelvins (-280° Fahrenheit), only slightly warmer than the boiling point of liquid nitrogen.

Titan has no appreciable magnetic field and therefore possesses no large liquid conducting core. It does, however, supply a small amount of charged

particles to Saturn's magnetosphere.

The southern hemisphere is somewhat brighter than the northern, perhaps as a result of seasonal effects.

OUTER SATELLITES

Of the three known outer satellites, Voyager 1 studied from a distance only Hyperion and Iapetus. Tiny Phoebe, in its retrograde (clockwise) orbit, will be studied by Voyager 2 in the summer of 1981. Hyperion and Iapetus are most likely composed of water ice, although their masses and densities are uncertain. Iapetus has one bright and one dark hemisphere. The dark side, which faces forward as Iapetus circles Saturn, reflects about one-fifth as much light as the trailing, bright side.

MAGNETOSPHERE

Although it is only about one-third the size of Jupiter's magnetosphere, Saturn's magnetosphere is still an enormous structure, extending nearly two million kilometers from the planet toward the Sun. The size of the magnetosphere fluctuates rhythmically as the flow of charged particles in the solar wind increases or decreases in intensity. The magnetosphere can be pushed inside Titan's orbit, so that at times the satellite finds itself outside of the magnetosphere altogether.

Charged particles in the planet's magnetosphere are dragged along by the magnetic field, circling the planet at Saturn's rotation rate of 10 hours, 39 minutes. These charged particles whiz by Titan at a dizzying rate of more than 200 kilometers (120 miles) per second. Titan leaves a motorboat-like wake in its orbital path.

Extending from the orbit of Titan inward to the orbit of Rhea, an enormous cloud of uncharged hydrogen atoms forms a doughnut-shaped torus of ultraviolet-emitting particles. Because of their neutrality, these atoms are not towed around by Saturn's magnetic field.

Close to the planet, Saturn's rings act as an effective shield or absorber of charged particles. The rings themselves are apparently substantially affected in this process, however, as evidenced by their "spokes" of fine particles and the lightning-like electrical discharges attributed to the rings.

Scientific Investigations

INVESTIGATION	SATURN ENCOUNTER OBJECTIVES
Imaging science	Planetary meteorology; satellite geology; ring structure and dynamics
Infrared interferometry	Atmospheric composition, thermal structure and dynamics; satellite surface composition and thermal properties; ring composition
Radio science	Atmospheric and ionospheric structure, constituents, and dynamics at Saturn and Titan; ring particle size
Ultraviolet spectroscopy	Upper atmospheric composition and structure; auroral processes; distribution of ions and neutral atoms in the Saturn system
Magnetic fields	Planetary magnetic field; magnetospheric structure
Plasma particles	Magnetospheric ion and electron distribution; solar wind interaction with Saturn; ions from satellites
Plasma waves	Plasma electron densities; wave-particle interactions; low-frequency wave emissions
Planetary radio astronomy	Polarization and spectra of radio-frequency emissions; plasma densities
Low-energy charged particles	Distribution, composition, and flow of energetic ions and electrons; satellite-energetic particle interactions
Cosmic ray particles	Distribution, composition, and flow of high-energy trapped nuclei; energetic electron spectra

“Notre voyageur connaissait merveilleusement les lois de la gravitation, et toutes les forces attractives et répulsives. Il s’en servait si à propos, que tantôt à l’aide d’un rayon de soleil, tantôt par la commodité d’une comète, il allait de globe en globe, lui et les siens, comme un oiseau voltige de branche en branche.”

“Our voyager knew marvelously the laws of gravitation, and all attractive and repulsive forces. He used them in such a timely way that, once with the help of a ray of sunshine, another time thanks to a cooperative comet, he went from globe to globe, he and his kin, as a bird flutters from branch to branch.”

VOLTAIRE—Micromégas, Histoire Philosophique, 1752.

42



National Aeronautics and Space Administration

Jet Propulsion Laboratory
California Institute of Technology
Pasadena, California

JPL 400-100 12/80

Transcriber's Notes

- Retained publication information from the printed edition: this eBook is public-domain in the country of publication.
- Silently corrected a few palpable typos.
- Moved captions nearer the relevant images; tweaked image references within captions accordingly.
- Added a Table of Contents.
- In the text versions only, text in italics is delimited by underscores.

*** END OF THE PROJECT GUTENBERG EBOOK VOYAGER 1
ENCOUNTERS SATURN ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.